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REVIEW

Circulating Biomarkers of Military-Specific Performance Resilience: A Systematic Review

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Abstract

Purpose of Review To systematically examine the available literature on circulating biomarkers of performance resilience in a military environment, with the goal of identifying the most promising circulating biomarkers.

Recent Findings The construct 'resilience' is hypothesized to play an important role in increasing Special Operations Forces' and other military personnel's capacity for withstanding exposure to various military-specific stressors. However, objectively measuring resilience is challenging. Some of the most important and well-studied circulating biomarkers that affect military-specific resilience are cortisol, dehydroepiandrosterone (sulfate) [DHEA(S)], noradrenaline, serotonin, neuropeptide-Y (NPY) and brain-derived neurotrophic factor. Despite growing evidence, the available knowledge is yet to be summarized and reviewed while considering the intensity and duration of military-specific stressors, military experience, and methodological differences between studies.

Summary Cortisol, Insulin-like growth factor-1 (IGF-1), NPY and DHEA(S) provide a physiological window into military-specific resilience. In general, individuals who exhibit a pronounced but controlled biomarker response to an acute stressor, combined with a quick recovery to baseline, demonstrate physiological flexibility that is associated with greater military-specific resilience. Future research will need to determine relative thresholds for the acute stressor-related change in circulating biomarkers and relative timing to stressor, to correctly interpret 'a pronounced but controlled biomarker response' and 'quick recovery to baseline'.

Keywords Trait resilience · State resilience · Hardiness · Toughness · Performance · Circulating biomarkers

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Introduction

Military personnel train and operate in volatile, uncertain, complex, and ambiguous environments where they are exposed to a variety of stressors. Military-specific stressors include a combination of physical exertion, variations in workload, sleep restriction, caloric restriction, variations in the operational environments, and emotional and psychological stressors. Despite being unavoidable features of military deployments, these stressors can negatively impact performance, health, safety, and mission success [1]. Furthermore, unrestricted, repeated, or chronic exposure to these stressors can contribute to excessive and generalized fear, hypervigilance, anxiety, the development of post-traumatic stress disorder, medical disorders including cardiovascular disease, and functional impairment [2]. Within Special Operations Forces [3–7] and other military units [8–12], the construct 'resilience' has systematically gained attention in the last

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decade, given that resilience is hypothesized to play an important role in increasing the capacity to withstand such challenges to homeostasis [9]. Varied definitions of resilience exist. For example, resilience has been defined as 'the process of negotiating, managing, and adapting to significant sources of stress or trauma' [9, 13], as 'a person's psychological ability to resist and adapt to significant stress, adversity and trauma' [10], and as 'the individual's ability to achieve, retain, or regain a level of physical or emotional health after illness or loss' [10]. These definitions highlight that resilience is sometimes described too broadly, rather than being recognized as a targeted response to a specific stressor or challenge [14]. In addition, the definitions also touch upon the ongoing debate over whether resilience should be viewed as a stable trait—a relatively enduring psychological disposition—or as a dynamic state that fluctuates depending on situational demands and stressor exposure.

From a psychophysiological perspective, this distinction is critical. Trait resilience reflects relatively stable interindividual differences that influence how stressors are appraised (i.e., as a threat or challenge), which in turn modulates physiological responses to stressors through activation of the sympathetic–adrenal–medullary system and the hypothalamic–pituitary–adrenal axis. In contrast, state resilience reflects transient adaptive responses within these same systems that support short-term recovery and performance under acute stress [15]. However, because resilience has been operationalized in diverse ways, its objective measurement remains challenging and often nonspecific and the discussion on whether (military-specific) resilience is rather a trait or a state remains ongoing.

The common concept shared by all definitions of resilience is 'the reaction to stressors'. Thus, an important step in objectifying resilience is to be able to quantify the impact of stressors. Since stressors impact both physiological and affective processes [8], ideal measures of stressor impact would include both of them. However, in the applied field, physiological measurements have several advantages compared to measuring affective processes. For example, no conscious input is needed from the individual being evaluated, eliminating the risk of bias and underreporting. As such, physiological screening and monitoring have been put forward as the main strategies to objectively measure stress and stressor reactivity, and, as such, resilience [1, 16]. These objective measures could be used to identify resilient individuals or those at risk of stress-related performance decrements. Establishing an individual's resilience level (i.e., assessing his/her targeted response to a specific stressor) prior to performance under severe stress or the development of clinical stress disorders [e.g., post-traumatic stress disorder (PTSD)] could facilitate focused preparation such as tailoring training to individual needs.

The response to a stressor is a highly conserved adaptive mechanism that enables individuals to cope with environmental challenges and ensure survival. When a potential threat or demand is perceived, the brain activates a coordinated psychophysiological cascade involving neural, endocrine, and immune systems to promote allostasis—the process of achieving stability through change [17]. This activation mobilizes energy, sharpens attention, and prepares the body for immediate action. While such responses are essential for short-term adaptation, repeated or prolonged activation can result in allostatic load, representing the cumulative wear and tear on physiological systems [17]. In military contexts, where individuals are repeatedly exposed to high-intensity and unpredictable stressors, an efficient and well-regulated response to a stressor is critical for maintaining performance, decision-making, and long-term health—key components of resilience [18].

Multiple physiological systems play a key role in the human response to a stressor, including the neuroendocrine, cardiac, and cognitive-behavioral systems [19]. Here, a stressor refers to any external or internal stimulus that challenges homeostasis, while stress refers to the body's physiological and psychological response to that challenge. Numerous hormones, neurotransmitters and neuropeptides are involved in the acute, sub-acute and chronic psychobiological responses to stressors. Due to the relative ease of sampling, peripherally circulating biomarkers, i.e., in the bloodstream, have been the focus of most of the recent studies [20]. Unfortunately, not all neurotransmitters are measurable in a peripheral blood sample. Nevertheless, differences in the functioning, balance, and interactions of peripherally circulating biomarkers underlie inter- and intra-individual variability in resilience [21]. As such, the focus on circulating biomarkers provides a thorough and insightful window into stressor reactivity [20] – a window that becomes increasingly applicable in the field (see the review of Vignesh et al. [22] for an overview). Some of the most important and well-studied circulating biomarkers that are associated with military-specific resilience are cortisol, dehydroepiandrosterone (sulfate) [DHEA(S)], neuropeptide-Y (NPY) and brain-derived neurotrophic factor (BDNF) [21].

In the presence of a stressor, the brain initiates a response that engages the immune and cardiovascular systems via neuroendocrine mechanisms, including the secretion of cortisol and DHEA [23]. The surge in cortisol is essential to mobilize and replenish energy stores, suppress non-essential anabolic activity, and increase arousal and cardiovascular tone [23, 24]. DHEA is a precursor for the synthesis of anabolic steroids and is co–released with cortisol from the adrenal cortex in response to a stressor. DHEA-S is the sulphated form of DHEA and the pool of DHEA-S serves as a reservoir for DHEA (i.e., only desulphated DHEA is biologically active). Concentrations of DHEA-S are much higher than concentrations of DHEA, partly because DHEA-S has longer half-life and lower clearance than DHEA [25]. As such, the increased concentration of DHEA-S during acute stress is much larger than for DHEA. The stressor-induced increase of DHEA-S accounts for the majority of the total increase of DHEA and DHEA-S together [26]. While the physiological consequences of DHEA are not completely understood, DHEA might counter the actions of cortisol as well as exert anti–oxidant and anti–inflammatory effects [27]. As a result, the ratio of DHEA-to-cortisol is often used to represent the balance between anabolic and catabolic processes [25]. Other neuroendocrine markers that have a protective role in the response to a stressor include NPY and BDNF. NPY is a pancreatic polypeptide that inhibits excessive activation of the response and possibly acts as an endogenous anxiety-reducing agent to buffer the effects of stress in the brain [23]. BDNF is also released both centrally and peripherally to protect neurons and mobilize energy resources during stress [23].

Research on circulating biomarkers of military-specific resilience has been reviewed before: Beckner et al. [20] summarized the key circulating biomarkers of physiological resilience by classifying them in neuroendocrine, anabolic and growth factor, and inflammatory domains; while Lee et al. [1] focused on physiological and biomechanical markers of heat or cold strain and how these may be used to ensure operational readiness in hot and cold environments. Both narrative reviews emphasized the large inter- and intra-individual variability in the reactivity of circulating biomarkers to stressors, and encouraged pursuit of an optimal individualized physiological monitoring method that can identify and propose an intervention for at-risk soldiers [1, 20]. However, as illustrated by the review of Beckner et al. [20], the different type of military stressors used, ranging from basic military training to elite forces selection, greatly influences the observed effects on circulating biomarkers of resilience. Indeed, the large inter- and intra-individual variability in the reactivity of circulating biomarkers to stressors are likely triggered by differences in intensity and duration of the stressor, and by the level of military experience [20]. Moreover, methodological differences between studies, such as differences in the timeframe of blood collections, preclude consistent results, thus making reproducibility of results challenging [20].

Therefore, to further advance this field of research, the available knowledge must be summarized and reviewed while considering the above-described variables, namely intensity of stressor, duration of stressor, military experience, and methodological differences between studies. Compared to the narrative reviews of Beckner et al. [20] and Lee et al. [1], a systematic review will provide a complete overview of the available literature and estimation of the impact of the above-described variables. The present systematic review aimed to include all research on circulating biomarkers of military-specific performance resilience that 1) stratified their cohort based on a performance resilience measure and 2) performed a statistical analysis that provides insight into the link between the resilience outcome and circulating biomarkers. To be parsimonious in a complex field of research, the authors decided to focus on the behavioral outcome of non-pathological military-specific resilience, i.e., the ability to perform under military-specific stressors and/or successfully complete a military-specific training and/or selection course (i.e., performance resilience). Also, in light of producing a clear overview of the available research and results, the different circulating biomarkers were classified based on the proposed classification by Beckner et al. [20]: 1) neuroendocrine, 2) anabolic and growth factors, and 3) inflammatory.

Method

In accordance with the updated "Preferred Reporting Items for Systematic review and Meta-analyses" (PRISMA) guidelines of 2020 [28], we undertook a systematic review. In addition, the protocol was also registered with PROSPERO (ID = CRD42024520591).

Eligibility Criteria

To establish the keywords for this systematic review, PICOS categories (Population, Intervention, Comparison, Outcome and Study design) were used (see Table 1). Studies were considered eligible for inclusion if they were analytical observational (i.e., Cohort studies, cross-sectional studies) or experimental [Clinical Trials (CTs), Controlled Clinical Trials (CCTs) or Randomized Controlled Trials (RCTs)] study designs [29]; evaluated the possible role of circulating biomarkers (i.e., we focused on blood-based biomarkers, but also included studies that determined saliva- and/or urine-based biomarkers that are also determinable in blood); and in the context of military-specific performance resilience (i.e., the ability to perform under military-specific stressors and/or successfully complete a military training and/or selection course).

Information Sources and Search Strategy

Three electronic databases, PubMed, EMBASE and PsycNET (until 06/08/2025), were searched. Medical subject heading (MeSH) terms, if available in PubMed, were used for a qualitative literature search. No limits were applied to the employed databases. See Table 1 for an overview of the keywords that were used.

Study Selection

Screening processes were conducted using Covidence (covidence.org), an online platform that supports meta-analysis collaboration between multiple researchers. Inclusion/exclusion of a study was decided with application of the PICOS criteria to the title, abstract, and/or full text of studies. Titles and abstracts were screened first by two reviewers (A.M and B.F; Level of agreement was 95%), conflicts were resolved by a third reviewer (J.V.C). Subsequently, full-text studies were retrieved if the citation was considered potentially eligible and relevant, and screened again by two reviewers (B.F and J.V.C; Level of agreement was 96%). The reference lists of included studies (i.e., backward screening), and studies that cited the included articles (i.e., forward screening) were then screened to make the search as complete as possible. The study-inclusion process is presented in Fig. 1.

Data Extraction

Data was extracted to evaluate the link between circulating biomarkers and the performance outcome of military-specific resilience, i.e., successfully completing a military training or selection course or upholding physical and/or cognitive performance throughout a military-specific stressor scenario. Extracted features on circulating biomarkers included: type of circulating biomarker (i.e., neuroendocrine, anabolic and growth factors and inflammatory), analysis method, and timing of sample collection in relation to the military-specific stressor. Extracted details on military-specific stressors included: type of stressor (e.g., training, selection or operation), duration, and intensity. Extracted details on resilience included: type of performance

Table 1 Number of hits on keywords and combined keywords in PubMed, EMBASE and PsycNET

Keywords	PubMed	EMBASE	PsycNET
	Hits (06/08/2025)	Hits (06/08/2025)	Hits (06/08/2025)
1) ((“Biomarkers” [MeSH] OR (“Biomarker”) OR (“Blood”) OR (“Hormon*”) OR (“Neurotransmitter”) OR (“Neuroendocrine”) OR (“Cortisol”) OR (“Epinephrine”) OR (“EPI”) OR (“Adrenaline”) OR (“Norepinephrine”) OR (“NE”) OR (“Noradrenaline”) OR (“Neuropeptide-Y”) OR (“NPY”) OR (“Brain Derived Neurotropic Factor”) OR (“BDNF”) OR (“Inflammatory”) OR (“Interleukin”) OR (“IL-6”) OR (“IL-1β”) OR (“IL-4”) OR (“IL-10”) OR (“Tumor Necrosis Factor”) OR (“TNF-α”) OR (“Anabolic”) OR (“Insulin-like Growth Fast-I”) OR (“IGF-I”) OR (“Testosterone”) OR (“Dehydroepiandrosterone”) OR (“Dhea”) OR (“Oestrogen”) OR (“Progesterone”) OR (“Conticotropin-releasing Hormone”) OR (“CRH”) OR (“Adrenocorticotropic Hormone”) OR (“ACTH”) OR (“Galanine”) OR (“Dopamine”) OR (“Serotonin”) OR (“Monoamine oxidase A”) OR (“Gamma-Aminobutyric Acid”) OR (“GABA”) OR (N-methyl-D-Aspartate OR (“NMDA”) OR (“Glutamate”))	7,781,793	10,257,009	400,503
2) ((“Resilience, Psychological” [MeSH] OR (“Emotional Adjustment” [MeSH]) OR (“Hardiness”) OR (“Readiness”) OR (“Resilien*”) OR (“Adaptation”) OR (“Adjustment”) OR (“Coping”) OR (“Cope”) OR (“Persistence”) OR (“Self-control”) OR (“Stress”))	2,196,515	2,875,160	772,308
3) ((“Resilience, Psychological” [MeSH]) OR (“Military Health” [MeSH]) OR (“Military”) OR (“Army”) OR (“Navy”) OR (“Air Force”) OR (“Space Force”) OR (“Coast Guard”) OR (“National Guard”) OR (“Armed Forces Personnel”) OR (“Special Forces”) OR (“Submariner”) OR (“Soldier*”) OR (“Marines”) OR (“Infantry”) OR (“Airmen”) OR (“Military Pilot”))	345,206	130,291	88,919
Combined keywords			
(1) AND (2)	546,564	868,584	71,167
(1) AND (3)	73,469	19,240	5243
(1) AND (2) AND (3)	7321	3682	2027

resilience outcome (e.g., success or change in physical or cognitive performance throughout a stressor scenario), link between circulating biomarker and resilience outcome (i.e., outcome). Other information that was extracted included: sample size and characteristics (e.g., years of service), geographic area where the data was collected, study objective, statistical analysis, and remarks. Missing data were not pursued in any form and were, if relevant, added to the quality assessment.

Quality Assessment

The methodological quality was assessed using the quantitative assessment tool 'QualSyst' by Kmet et al. [30]. QualSyst contains 14 items (see Table 2) that are scored depending on the degree to which the specific criteria were met (yes = 2, partial = 1, no = 0). Items not applicable to a particular study design were marked 'NA' and excluded from the calculation of the summary score. A summary score was calculated for each article by summing the total score obtained across relevant items and dividing it by the total possible score. Two reviewers (Y.A and J.V.C) independently performed quality assessments, and disagreements were solved by consensus or by a third reviewer (B.F). A score of ≥ 75% indicated strong quality, a score of 55–75% indicated moderate quality, and a score ≤ 55% indicated weak quality.

Results

Study Selection

The systematic literature search yielded 10,628 unique articles. Via title and abstract screening, 10463 studies were excluded. Subsequently, after full text screening, 27 studies were included. A forward search (i.e., assessing the citations of the included articles) and backward search (i.e., assessing the reference lists of the included articles) provided 13 additional papers, bringing the total of included papers to 40. The level of agreement between the two authors that participated in the title and abstract screening and in the full text screening was, respectively, 95% and 96%. The full study selection process is presented in Fig. 1.

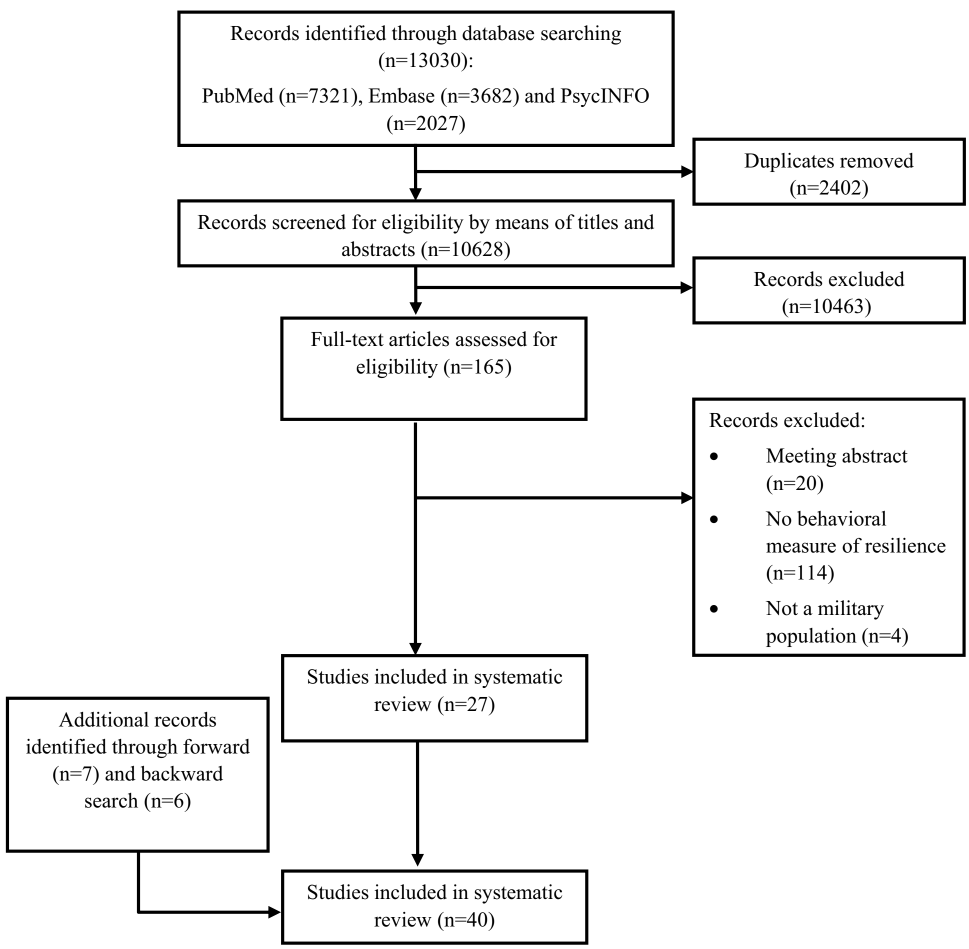

Fig. 1 PRISMA flowchart describing the process of obtaining the research articles ($n = 40$) included in this systematic review

Table 2 Quality assessment 'Qualsyst'

Study	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Rating
Binsch et al. 2017	2	1	N/A	1	N/A	N/A	N/A	1	1	2	2	2	2	2	**80**
Conkright et al. 2021	2	2	N/A	2	N/A	N/A	N/A	2	2	2	1	1	2	2	**90**
Cuddy et al. 2013	2	1	N/A	0	N/A	N/A	N/A	1	1	2	1	0	1	2	**55**
Farina et al. 2019	2	1	N/A	1	N/A	N/A	N/A	1	2	2	2	2	2	2	**85**
Farina et al. 2025	2	2	N/A	1	N/A	N/A	N/A	1	2	2	2	2	2	2	**90**
Forse et al. 2024	2	1	N/A	2	N/A	N/A	N/A	2	2	2	2	1	2	2	**90**
Gepner et al. 2019	2	1	N/A	1	N/A	N/A	N/A	2	1	2	2	0	1	2	**70**
Goring et al. 2025	2	2	N/A	2	N/A	N/A	N/A	1	2	2	1	2	2	2	**90**
Hamarsland et al. 2018	1	2	N/A	0	N/A	N/A	N/A	2	1	1	2	0	2	2	**65**
Iyer et al. 1994	1	1	N/A	0	N/A	N/A	N/A	1	2	1	2	0	2	2	**60**
Jenz et al. 2021	2	2	N/A	1	N/A	N/A	N/A	1	2	2	2	0	2	2	**80**
Kalns et al. 2011	2	1	N/A	1	N/A	N/A	N/A	1	2	2	2	1	2	2	**80**
Kargl et al. 2024	2	1	N/A	1	N/A	N/A	N/A	1	2	2	2	1	2	2	**80**
Ledford et al. 2020	2	1	N/A	0	N/A	N/A	N/A	1	2	1	2	2	2	2	**75**
Ledford et al. 2022	1	2	N/A	0	N/A	N/A	N/A	1	2	2	2	2	2	2	**80**
Leino et al. 1999	2	2	N/A	1	N/A	N/A	N/A	1	2	1	2	0	2	2	**75**
Martin et al. 2025	2	2	N/A	1	N/A	N/A	N/A	1	2	2	2	1	2	2	**85**
McFadden et al. 2024	1	1	N/A	0	N/A	N/A	N/A	1	2	2	2	0	1	2	**60**
Morgan III et al. 2000	1	2	N/A	1	N/A	N/A	N/A	1	2	2	1	1	1	2	**70**
Morgan III et al. 2001	2	2	N/A	2	N/A	N/A	N/A	1	2	1	2	2	1	2	**85**
Morgan III et al. 2004	1	2	N/A	1	N/A	N/A	N/A	1	1	2	2	2	1	0	**65**
Morgan III et al. 2009	2	2	N/A	1	N/A	N/A	N/A	2	2	1	2	2	2	0	**80**
Nieto Jimenez et al. 2018	2	2	N/A	1	N/A	N/A	N/A	2	2	1	2	0	2	2	**80**
Nindl et al. 2007	1	2	N/A	1	N/A	N/A	N/A	2	2	1	2	0	2	2	**75**
Nindl et al. 2012	1	2	N/A	1	N/A	N/A	N/A	2	2	2	2	0	2	2	**80**
Ojanen et al. 2018a	1	2	N/A	1	N/A	N/A	N/A	2	2	1	2	0	2	2	**75**
Ojanen et al. 2018b	1	2	N/A	1	N/A	N/A	N/A	2	2	1	2	0	2	2	**75**
Pihlainen et al. 2020	1	2	N/A	2	N/A	N/A	N/A	2	2	1	2	0	2	2	**80**
Ponce et al. 2023	2	2	N/A	1	N/A	N/A	N/A	2	2	2	2	0	2	2	**85**
Ponce et al. 2025	2	2	N/A	1	N/A	N/A	N/A	1	2	1	2	1	1	2	**75**
Sekel et al. 2023	2	2	N/A	2	N/A	N/A	N/A	2	2	2	2	0	2	2	**90**
Shia et al. 2015	2	2	N/A	1	N/A	N/A	N/A	2	1	0	1	0	1	2	**60**
Shumway et al. 2020	2	2	N/A	0	N/A	N/A	N/A	1	1	1	2	1	2	2	**70**
Stein et al. 2023	2	1	N/A	1	N/A	N/A	N/A	2	2	2	1	2	2	2	**85**
Szivak et al. 2018	2	2	N/A	1	N/A	N/A	N/A	2	1	1	2	2	2	2	**85**
Tait et al. 2022	2	1	N/A	1	N/A	N/A	N/A	1	2	2	2	2	2	2	**85**
Vaara et al. 2020	2	1	N/A	0	N/A	N/A	N/A	1	2	2	2	2	2	2	**80**
Vaernes et al. 1982	2	2	N/A	0	N/A	N/A	N/A	2	1	0	2	0	2	1	**60**
Varanoske et al. 2022	2	2	N/A	1	N/A	N/A	N/A	2	2	2	2	1	2	2	**90**
Vrijkotte et al. 2017	1	2	N/A	2	N/A	N/A	N/A	1	1	2	2	1	2	2	**80**

Table 2 (continued)

A = Question described specifically related to the role of circulating biomarkers in performance resilience?, B = Appropriate study design? (i.e., included multiple blood sample collections? Performed blood sampling or only urine and/or saliva sampling?), C = Appropriate subject selection?, D = Characteristics described? (i.e., age, sex, military experience, physical fitness level?), E = Random allocation?, F = Researchers blinded?, G = Subjects blinded?, H = Outcome measures well defined and robust to bias? (i.e., was the performance resilience measure well described? How was decided whether someone succeeds or not?), I = Sample size appropriate?, J = Analytic methods well described? (i.e., included statistical methods and outcomes explained in detail?), K = Estimate of variance reported? (i.e., details provided in table/text or only visual details are provided in figure?), L = Controlled for confounding? (i.e., described in statistical analysis?), M = Results reported in detail? (i.e., if possible, a correlational analysis was performed with both absolute and change circulating biomarker scores?), N = Conclusion supported by results?. 2 = yes, 1 = partial, 0 = no, N/A = Not Applicable. Quality scores: Strong = ≥ 75%, Moderate = 55% > 75%, Weak = ≤ 55%.

Quality Assessment

Quality assessment of the 40 included studies with the QualSyst-tool determined that 30 studies are of strong quality, 9 studies of moderate quality and 1 study of low quality (see Table 2). Regarding the studies that were evaluated as moderate and low quality studies, the main items that resulted in the moderate quality score were 1) the description of the characteristics of the participants, 2) controlling for confounding variables, 3) the description of the statistical analysis and 4) the inclusion of a performance resilience measure that is robust to bias.

Study and Sample Characteristics

All relevant information regarding study characteristics, employed military-specific stressors, performance resilience outcome and evaluated circulating biomarkers can be found in Table 3 and 4. The total population of all articles was 5058 participants, with only 484 of them being women. Six different studies included women [31–36]. The included study's population ranged from 11 [37] to 1006 [36] participants and, while most studies have been performed in the USA [87% (n=4407) of the pooled participants were included in USA-based studies], the total included population in the current systematic review represents an international population [13% (n=651) of the pooled participants were included in non-USA-based studies] with studies conducted in the Netherlands [38], Belgium [39], Norway [40, 41], Israel [32, 42], Finland [11, 43–46], Brazil [47, 48], Australia [33] and Chile [49]. The average age of the participants ranged from 19 to 30 years. Included participants were all operational soldiers, e.g., Special Forces soldiers, active-duty U.S. Army Soldiers, or candidates to become operational in a specific (Special Forces) unit, e.g., Elite combat unit of the Israel Defense Forces or Navy Sea, Air, and Land Teams (SEAL) candidates. In the studies that reported the number of years of service of the included participants [50–53], this number was varied from 2.5 [53] to 7 years [50]. Besides certain military experience, almost all studies also reported that participants were screened prior to being included and had to demonstrate high physical and/or mental fitness (e.g., 'Possess an Armed Services Vocational Aptitude Battery General Technical score of 110 or higher' [54] or 'Specifically screened for their athleticism, mental stability, and intelligence' [23]), and/or successfully completing a certain preparatory course (e.g., 'Passed the selection and Air Mobile Brigade fitness test' [38]).

Military-Specific Stressors

Regarding military-specific stressors that are included in the studies of the present systematic review, a distinction can be made between studies that evaluated the role of circulating biomarkers in performance resilience during prolonged military-specific stressors (e.g., multi-day training or selection course), and studies that evaluated their role during acute military-specific stressors (e.g., physically exhausting treadmill protocol, mentally taxing dive). Both stress situations are often encountered throughout a military career, and it is therefore of value to assess the possible role of circulating biomarkers in performance resilience for both these scenarios.

Prolonged Military-Specific Stressor Studies

Of the 40 included studies, 34 included a prolonged stressor in their study design (see Table 3). Twenty of them monitored circulating biomarkers in military throughout a military training program/course [4, 6, 8, 31, 32, 34, 35, 38, 39, 42–44, 47, 50–52, 55–57], another 13 studies focused on circulating biomarkers time dynamics in a pass/fail selection course [11, 23, 33, 36, 37, 40, 48, 54, 58–63], and one study evaluated the role of circulating biomarkers on the impact of a military operation [45]. The observed stressors lasted from 24 h [63] to two years [62], with most studies focusing on a stressor lasting from a few days to a month. Regarding the intensity of the stressors, there was a clear similarity between the studies, as most stressors included intense physical and mental exertion, as well as sleep restriction in a stressful environment. For example: the stressor included in the study of Morgan III et al. [55] was a Combat Diver Qualification course that involves drown-proofing exercises, open ocean swimming/navigation and nocturnal underwater navigation training; Szivak et al. [6] monitored participants throughout a Survival, Evasion, Resistance, and

Table 3 Overview of the military-specific stressors that were employed in the included studies and the circulating biomarker(s) that were evaluated

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Prolonged military-specific stressor studies								
Binsch et al. [38] (experiment 2)	29 (no info on M or F)	Netherlands	Investigate the relationship between cortisol and optimism, and their association to perseverance, that is, to attrition in a military basic training setting	Weeks 6 and 7 of the 24-week Air Mobile Brigade training program	The complete Air Mobile Brigade training program is 24 weeks	**Week 6**: Marksmanship training (low pressure) **Week 7**: field exercise (challenging)	**<u>Saliva sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -** **Inflammatory: -**	Radioimmunoassay method
Conkright et al. [31]	69 (54 M & 15 F)	USA	Compare physical performance and hormonal trajectories in men and women undergoing SMOS lasting 5 days	SMOS protocol	5 days	Frequent cognitive testing, marksmanship, physical exertion, energy deficit, and sleep restriction	**<u>Blood sample</u>** **Neuroendocrine:** • Cortisol • BDNF **Anabolic & growth factor:** • IGF-1 • GH **Inflammatory: -**	Standard ELISA or magnetic bead-based procedures (BDNF only)
Cuddy et al. [37]	11 (no info on M or F)	USA	Identify whether return and/or successful SOF candidates demonstrated differences in cortisol responses compared to non-selected and/or first-time attendees	United States Air Force Special Tactics Officer selection	5 days	Including: running, swimming, calisthenics, ruck marching, water skill sessions, leadership reaction courses, and 'Monster Mash'	**<u>Saliva sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -Inflammatory: -**	Immunoassay
Farina et al. [54]	800 (M)	USA	Establish characteristics associated with successful performance in mentally and physically stressful environments	SFAS course	19–20 days	Extensive cognitive (intelligence, team problems solving tasks) and physical challenges (timed loaded road marches)	**<u>Blood sample</u>** **Neuroendocrine:** • Cortisol • Adrenaline • Noradrenaline **Anabolic & growth factor:** • DHEA-S • Testosterone • SHBG **Inflammatory:** • CRP	Immunoassay using a chemiluminescent detection method ELISA (adrenaline & noradrenaline)

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Farina et al. [62]	233 (M)	USA	To determine whether psychological, physiological, or nutritional measures at baseline, or changes in these measures during SFAS, would predict likelihood of graduation from SFQC	SFAS and SFQC course	SFAS = 19–20 days SFQC = 2.2 ± 0.49 y	SFAS = see above SFQC = Orientation to Special Forces culture and history; military occupational specialty training; SERE school; tactical skills training; and foreign language training	<u>**Blood sample**</u> **Neuroendocrine:** • Cortisol • BDNF • NPY • Prolactin • Parathyroid hormone • Adrenaline • Noradrenaline **Anabolic & growth factor:** • DHEA-S • Testosterone • SHBG • Insulin • Ghrelin • Iron • Glycated hemoglobin **Inflammatory:** • IL-6 • High sensitivity CRP • Ferritin	Immunoassay, ELISAs and radioimmunoassay
Forse et al. [36]	1006 (79.5% M)	USA	To examine the predictors of attrition in male and female candidates undergoing a 10-week early career military training program	US Marine Corps Officer Candidates School	10 weeks	A unique training pipeline in which candidates are typically older in age and with higher education levels due to the requirement that candidates be working toward or have completed a college degree	<u>**Blood sample**</u> **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • DHEA **Inflammatory: -**	ELISAs

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Gepner et al. [42]	20 (M)	Israel	Examine the association between markers of inflammation and neurotrophin concentrations and marksmanship and cognitive function following a five-day field exercise simulating a sustained military operation	Intense field training simulating a sustained military operation	5 days	~27.8 km navigated/day, 200 m gauntlet run, and marksmanship	**Blood sample** **Neuroendocrine:** • BDNF • GFAP **Anabolic & growth factor: -** **Inflammatory:** • IL-10 • TNF-α	Multiplex assay (IL-10 and TNF-α) ELISA (BDNF and GFAP)
Hamarsland et al. [40]	15 (no info on M or F)	Norway	Investigate the effect of an extremely demanding 1-wk military training course on physical performance, body composition and blood biomarkers among apprentices to the Norwegian Naval Special Operations Command	Annual selection course to join the Norwegian Naval Special Forces	First 6 weeks of the selection course	Heavy physical activity and sleep restriction in a stressful environment for 3 weeks + hell week + 2-week recovery period	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • IGF-1 • Testosterone • SHBG • TSH • T3 • Thyroxine • IGFBP-3 **Inflammatory:** • CRP • CK	Analyzed at Fürst (Oslo, Norway) and at The Hormone Laboratory (Oslo, Norway)
Jenz et al. [8]	33 (M)	USA	Whether a panel of bio-and affective markers can differentiate between subjects that eventually graduate from their Course Of Initial Entry versus those who did not	Battlefield Airmen Preparatory Course prior to their Course Of Initial Entry	52 days	Consists of routine physical exercise and prolonged and intense physical and mental training	**Blood sample** **Neuroendocrine:** • Cortisol • Noradrenaline • NPY • Orexin • 5HT **Anabolic & growth factor:** • Testosterone • DHEA-S **Inflammatory:** • CRP • CK • ALOX15B • STAT1 **Multiple other circulating biomarkers**	ELISA

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Kalns et al. [58]	122 (no info on M or F)	USA	To develop a statistical model that predicts the likelihood of success or failure of military training candidates using tests administered before initial skill training as inputs	U.S. Air Force Tactical Air Control Party training	No info	No info Tactical Air Control Party specialists, provide direct close air-support firepower and advise Army combat commanders	<u>**Saliva sample**</u> **Neuroendocrine: -** **Anabolic & growth factor: -** **Inflammatory: -** **Other:** • Ratio heptapeptide ESPSLIA/GGHPPPP	Spectrometry
Kargl et al. [35]	163 (101 M & 62 F)	USA	Cytokine and oxidative stress marker concentrations were correlated with performance testing outcomes to determine whether these variables are associated	US Marine Corps Officer Candidates School (	10 weeks	Candidates complete a variety of physical events such as obstacle courses, ruck marches and physical fitness tests	<u>**Blood sample**</u> **Neuroendocrine: -** **Anabolic & growth factor: -** **Inflammatory:** • TNF-α • IL-6 • IL-8 • IL-10 • CRP • IFN-γ	ELISA (CRP) Simple Plex Human Multianalyte Assay kit (IL-6, IL-10, IL-8, TNF-α, and IFN-γ)
Ledford et al. [59]	116 (no info on M or F)	USA	To utilize biomarkers indicative of biological stress to elucidate the biological signatures of physiological resilience	First Phase of the Basic Underwater Demolition/SEAL course	54 days	Extensive physical exertion and continuous haranguing by instructors	<u>**Blood sample**</u> **Neuroendocrine:** • Cortisol • BDNF **Anabolic & growth factor:** • DHEA **Inflammatory: -**	ELISA
Ledford et al. [23]	380 (no info on M or F)	USA	To identify growth patterns in physiological indicators (i.e., circulating blood biomarkers) during Basic Underwater Demolition/SEAL course	Complete Basic Underwater Demolition/SEAL course	28 weeks	This accession and selection process is designed to ensure that students meet the minimum mental and physical toughness threshold	<u>**Blood sample**</u> **Neuroendocrine:** • Cortisol • BDNF • NPY **Anabolic & growth factor:** • DHEA **Inflammatory: -**	ELISA

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating bio-marker analysis method
Martin et al. [63]	48 (M)	USA	To identify physical, physiological, and psychological differences between finishers and nonfinishers	The Naval Special Warfare screener	24 h	Constant stress is being imposed on the candidates. Some of these stressors include swimming, obstacle courses, running, calisthenics, boat drills, cold and sleep deprivation	**Blood sample** **Neuroendocrine:** • Cortisol • BDNF • NPY • ACTH • Adrenaline **Anabolic & growth factor:** • IGF-1 • DHEA • Testosterone • GH **Inflammatory:** • IL-6 • IL-1β • IL-10 • TNF-α **Saliva sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -** **Inflammatory: -**	ELISAs
McFadden et al. [34]	196 (97 M & 99 F)	USA	To evaluate training demands and characteristics that were associated with performance outcomes during the 13-week United States Marine Corps recruit training pro-gram	United States Marine Corps recruit training	13 weeks	Throughout training, recruits experience high internal and external workloads coupled with suboptimal sleep and increased stressors	**Saliva sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -** **Inflammatory: -**	ELISA
Morgan III et al. [4]	70 (M)	USA	To assess plasma NPY immunoreactivity in healthy soldiers participating in high intensity military training	Army survival training course	72 h	A captivity experience, each subject was subjected to intense and uncontrollable stress, and each attempted to avoid exploitation by their captors	**Blood sample** **Neuroendocrine:** • Cortisol • NPY **Anabolic & growth factor: -** **Inflammatory: -** **Saliva sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -** **Inflammatory: -**	Radioimmunoassay

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Morgan III et al. [50]	44 (M), subset from Morgan III et al. [4]	USA	To further explore the idea that differences in the neurobiological responses of individuals exposed to threat are significantly related to the concomitant psychological and behavioral indices	Army survival training course	72 h	A captivity experience, each subject was subjected to intense and uncontrollable stress, and each attempted to avoid exploitation by their captors	**<u>Blood sample</u>** **Neuroendocrine:** • Cortisol • Adrenaline • Noradrenaline • NPY **Anabolic & growth factor: -** **Inflammatory: -** **<u>Saliva sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:—Inflammatory: -**	Liquid chromatography (noradrenaline and adrenaline) NPY and cortisol (see Morgan III et al. [4])
Morgan III et al. [51]	25 (no info on M or F)	USA	To investigate plasma DHEA-S and cortisol levels, psychological symptoms of dissociation, and military performance	Army survival training course	72 h	Includes interrogations and problem-solving dilemmas designed to test the trainees' ability to use the information they learned during the didactic phase	**<u>Blood sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • DHEA-S **Inflammatory: -** **<u>Saliva sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -** **Inflammatory: -**	Radioimmunoassay
Morgan III et al. [55]	41 (M)	USA	To assess whether DHEA and DHEA-S would predict superior performance in soldiers participating in a nocturnal underwater navigation task, while at the Combat Diver Qualification Course	Combat Diver Qualification Course	1 month	Drown-proofing exercises, open ocean swimming/navigation, nocturnal underwater navigation training	**<u>Blood sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • DHEA • DHEA-S **Inflammatory: -** **<u>Saliva sample</u>** **Neuroendocrine:** • Cortisol **Anabolic & growth factor: -** **Inflammatory: -**	Radioimmunoassay

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Nindl et al. [56]	50 (M)	USA	To examine the relationship between changes in body composition and somatotrophic hormones after an extended energy deficit with deliberate energy restriction	U.S. Army Ranger training course	62 days	Involves four phases, each containing a period of normal feeding during field instruction, followed by 7–10 d of underfeeding during graded patrolling activities	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • IGF-1 • Testosterone **Inflammatory: -**	Radioimmunoassay
Nindl et al. [32]	122 (29 M & 93 F)	Israel	To assess the influence of sex and initial fitness level on the IGF-1 and inflammatory cytokine response to gender-integrated Israeli Defense Forces Basic Combat Training	Israeli Defense Forces Basic Combat Training	4 months	Includes marching, running, various training exercises, and stationary standing activities	**Blood sample** **Neuroendocrine: -** **Anabolic & growth factor:** • IGF-1 (total and free) • IGFBPs 1–6 **Inflammatory:** • TNF-α • IL-6 • IL-1β	2-side immunoradiometric assay (IGF-1), enzyme-linked immunosorbent assay (free IGF-1), Luminex xMAP technology (IGFBPs 1–6), enzyme-linked immunosorbent assay (IL-6)
Ojanen et al. [43]	49 (M)	Finland	To investigate the changes in hormonal and immunological values and body composition during a prolonged Military Field Training and to find out how warfighters' physical condition influences these changes	Military Field Training	22 days	**Phase 1**: Shooting exercises **Phase 2**: Practice moving from base to attack position **Phase 3**: Large-scale military exercise	**Blood sample** **Neuroendocrine: -** **Anabolic & growth factor:** • IGF-1 • Leptin **Inflammatory:** • TNF-α • IL-6 • CK	Immunoassay system and ELISA (leptin)
Ojanen et al. [44]	49 (M) same cohort as in Ojanen et al. [43]	Finland	To investigate the changes in body composition, upper and lower body strength, serum hormone (testosterone, cortisol, SHBG, IGF-1) concentrations, and shooting accuracy during a prolonged Military Field Training	Military Field Training	22 days	**Phase 1**: Shooting exercises **Phase 2**: Practice moving from base to attack position **Phase 3**: Large-scale military exercise	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • IGF-1 • Testosterone • SHBG **Inflammatory: -**	No info

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Pihlainen et al. [45]	49 (M)	Finland	To investigate differences in training responses and adaptations of endurance performance during combined strength and endurance training in a six-month crisis management operation in the Middle East	Crisis management operation in the Middle East	6 months	Operative duties including typical military tasks: • Patrolling and observing outside the military base, • Maintenance and headquarter duties inside the base	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • IGF-1 • Testosterone • SHBG **Inflammatory: -**	Chemiluminescent enzyme immunoassay kits
Ponce et al. [47]	42 (M)	Brazil	To identify possible correlations among physical performance, cognitive alertness, and hormone levels	Search and Rescue Field Military Exercise	9 days	Activities that simulate rescues under adverse conditions (high intensity and sleep restriction)	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • IGF-1 • GH • Testosterone **Inflammatory: -**	Chemiluminescence
Ponce et al. [48]	35 (M)	Brazil	To compare biomarker behavior between those who completed a 259 days high-intensity training course (finalists) and those who withdrew (dropouts)	Amphibious Commandos Specialisation Course Six operational missions	Each mission lasted 7–12 days	Operate in high-risk combat scenarios, including area mapping, rescues, strategic takeovers and amphibious missions	**Blood sample** **Neuroendocrine: -** **Anabolic & growth factor:** • Albumin • Creatinine • Urea • Estimated glomerular filtration rate • Electrolytes **Inflammatory:** • CK • Lactate dehydrogenase • Aspartate aminotransferase • Alanine aminotransferase • Gamma-glutamyl transferase	The dry chemistry method, the modified kinetic method and the pyruvate-lactate method

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Sekel et al. [52]	48 (M)	USA	To determine the impact of a 48-h SMOS on tactical adaptive decision making, and the influence of select psychological, physical performance, cognitive, and physiological outcome measures on decision making	SMOS	48 h	Involved sleep and caloric restriction and physical work, on military tactical adaptive decision making	**Blood sample** **Neuroendocrine:** • Cortisol • BDNF • NPY **Anabolic & growth factor:** • IGF-1 **Inflammatory: -**	ELISA
Shumway et al. [60]	Experiment 1 (n=12; no info on M or F) Experiment 2 (n=9; no info on M or F)	USA	To assess for biomarkers indicative of passing intense physical training and establishing normative values within the tactical athlete population	**Exp 1** Combat Control Operators Course at Keesler AFB **Exp 2** Combat Control School at Pope Field—Stress Inoculation Training	**Exp 1** No info on duration **Exp 2** 3 days	**Exp 1** Daily 2 h of exercise followed by 8 h of academics **Exp 2** A grueling pass/fail training event that involves sleep deprivation, mental and physical stress	**Blood sample** **Neuroendocrine: -** **Anabolic & growth factor:** **Inflammatory: -** • CK • Creatinine **Multiple other circulating biomarkers**	Siemens Dimension Vista 1500 machine No further info is provided
Stein et al. [61]	761 (M)	USA	To compare pre-SFAS blood metabolomes of soldiers selected during SFAS versus those not selected, and explore the relationships between the metabolome, physical performance, and diet quality	SFAS course	19–20 days	Candidates experience a multi-stressor environment with real world occupational consequences with limited information on upcoming events. Half of all the candidates fail the SFAS	**Blood sample** **Neuroendocrine: -** **Anabolic & growth factor: -** **Inflammatory: -** **Other:** • Metabolomics profiling	Mass spectrometry

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Szivak et al. [6]	20 (M)	USA	To expand upon prior work in the SERE population and further explore relationships among fitness level, physical performance and neuroendocrine responses	SERE Training	2 weeks	Captivity stressors (environmental extremes, sleep deprivation, psychological stress, food deprivation, and physical demands)	**Blood sample** **Neuroendocrine:** • Cortisol • Adrenaline • Noradrenaline • Dopamine • NPY **Anabolic & growth factor:** • Testosterone **Inflammatory: -**	High Performance Liquid Chromatography ((nor)adrenaline and dopamine) Radioimmunoassay (cortisol, testosterone and NPY)
Tait et al. [33]	46 (37 M & 9 F)	Australia	To determine the individual risk factors for training delays/ discharge during a Basic Military Training course	Basic Military Training	12 weeks	Involved physically and cognitively demanding activities and simulations	**Saliva sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • Testosterone **Inflammatory: -**	ELISA
Vaara et al. [11]	69 (no info on M or F)	Finland	To investigate if baseline physical fitness, body composition, hormonal and psychological factors could predict dropout from a 10-day intense winter military survival training	Winter survival training	10 days	Skiing on average 19.3 km/day with a 35–40 kg load and survival training. Also, sleep restriction, energy deficit, and continuous uncertainty and enemy evasion	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • IGF-1 • Testosterone • SHBG • DHEA • DHEA-S **Inflammatory:** • CRP	Immunoassay analyzer
Varanoske et al. [57]	68 (M)	USA	To determine changes in intestinal barrier and blood brain barrier permeability during stressful military survival training and identify relationships between those changes and markers of stress, inflammation, cognitive performance, and mood state	SERE Training	18 days	Phase 1: 10-day classroom training Phase 2: 2.5-day survival skills training Phase 3: 2.5-day evasion training Phase 4: 2.5-day captivity training	**Blood sample** **Neuroendocrine:** • Cortisol • Adrenaline • Noradrenaline **Anabolic & growth factor:** • DHEA-S **Inflammatory:** • IL-6 • High-sensitivity CRP **Other:** • Lipopolysaccharide-binding protein • S100 calcium-binding protein	ELISA

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Vrijkotte et al. [39]	24 (M)	Belgium	To investigate whether physiological, neuropsychological and/or subjective data could provide information about the ability to complete an advanced military training course	An advanced military training course	12 weeks	8 weeks paratrooper training and 4 weeks specific commando training	**<u>Blood sample</u>** **Neuroendocrine:** • Free cortisol • ACTH **Anabolic & growth factor: -** **Inflammatory: -**	No info
Acute military-specific stressor studies								
Goring et al. [53]	115 (M)	USA	To examine the relative utility of 12 salivary analytes for predicting cognitive and physical performance outcomes	6 different study events	No info	Participants engaged in a series of tasks that demanded, moving, shooting, communicating, navigating, and sustaining readiness	**<u>Saliva sample</u>** **Neuroendocrine:** • Cortisol • Alpha amylase **Anabolic & growth factor:** • Estradiol • DHEA • Testosterone • Creatinine • Lactate • Arginine • Carnitine • Carnosine • Histidine **Inflammatory:** • Urocanic acid	ELISAs
Iyer et al. [64]	48 (no info on M or F)	No info	To assess the catecholamine response of flight cadets during mid-term tests (49th sortie)	Missions performed during mid-term tests (49th sortie)	40 min	Consisted of 10 maneuvers such as: check procedure, taxiing, spin and recovery loop and roll left and right	**<u>Urine sample</u>** **Neuroendocrine:** • Catecholamine • Adrenaline • Noradrenaline **Anabolic & growth factor: -** **Inflammatory: -**	Fluorometric method
Leino et al. [46]	35 (M)	Finland	To evaluate neuroendocrine responses associated with stressful military flying and correlate these changes to flight performance	Instrument flight	40 min	3 non-precision approaches with a piston-engined primary trainer	**<u>Blood sample</u>** **Neuroendocrine:** • Cortisol • ACTH • β-endorphin • Prolactin • Adrenaline • Noradrenaline **Anabolic & growth factor: -** **Inflammatory: -**	Chemiluminometric immunoassay (prolactin) Radioimmunoassay (β-endorphin, ACTH, cortisol) Liquid chromatography (adrenaline and noradrenaline)

Table 3 (continued)

Study	Sample	Geographic area	Study objective	Military-specific stressor	Duration of stressor	Intensity of stressor	Evaluated circulating biomarker(s)	Circulating biomarker analysis method
Nieto Jimenez et al. [49]	42 (M)	Chile	To identify the endocrine, hematological, cardiovascular, and immunological changes caused by hypothermia and to associate these variables with body composition and physical fitness	Cold water plunge	Determined by individual volitional limits, ranging from 50 to 75 min	Soldiers remained in cold water (10.6 °C) wearing only undergarments	**Blood sample** **Neuroendocrine:** • Cortisol • ACTH **Anabolic & growth factor:** • Free testosterone • Total testosterone • TSH **Inflammatory: -**	Chemiluminescence and radioimmunoassay (free testosterone)
Shia et al. [65]	17 (M)	USA	To replicate the findings that decreased cortisol and increased DHEA-S correlate with increased performance outcomes in stressful military environments	Modified-Astrand Treadmill protocol	48–72 min	Treadmill protocol with 24 stages of 2 to 3 min. Speed remained constant (3.2 mph) and the incline increased to a grade that ranged between 2 and 16%	**Blood sample** **Neuroendocrine:** • Cortisol **Anabolic & growth factor:** • DHEA-S **Inflammatory: -**	Radioimmunoassay
Vaernes et al. [41]	29 (no info on M or F)	Norway	To determine whether cortisol correlates with specific Defense Mechanism Test variables, and whether this hormone correlates with the individual reduction on the performance tests	Dive in the pressure chamber	52 min	Dive with 10 min of bottom time at 60 m sea water deep, no physical work load	**Blood sample** **Neuroendocrine:** • Cortisol • Prolactin **Anabolic & growth factor:** • GH • Testosterone **Inflammatory: -** **Urinary sample** **Neuroendocrine:** • Catecholamines in urinary sample **Anabolic & growth factor: -** **Inflammatory: -**	Fluorometry (Catecholamines) Radioimmunoassays (GH, prolactine, cortisol and testosterone)

Abbreviations-legend: Serotonin (5HT), Adrenocorticotropic hormone (ACTH), Arachidonate 12-lipoxygenase (ALOX15B), Brain-derived neurotrophic factor (BDNF), Creatine kinase (CK), C-reactive protein (CRP), Dehydroepiandrosterone (DHEA), Dehydroepiandrosterone sulfate (DHEA-S), Dehydroepiandrosterone (sulfate) [DHEA(S)], Enzyme-Linked Immunosorbent Assay (ELISA), Female (F), Glial fibrillary acidic protein (GFAP), Growth hormone (GH), Interferon gamma (IFN-γ), Insulin-like growth factor-1 (IGF-1), Insulin-like growth factor binding protein (IGFBP), Interleukin (IL), Male (M), Neuropeptide Y (NPY), Navy, Sea, Air, and Land Teams (SEAL), Survival, Evasion, Resistance, and Escape (SERE), Special Forces Assessment and Selection (SFAS), Special Forces Qualification Course (SFQC), Sex hormone binding globulin (SHBG), Simulated military operational stress (SMOS), Special Operations Forces (SOF), Signal transducer and activator of transcription 1 (STAT1), Triiodothyronine (T3), Tumor Necrosis Factor alpha (TNF-α), Thyroid stimulating hormone (TSH)

Table 4 Overview of the included studies' reported results on the role of circulating biomarkers in military-specific performance resilience

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Prolonged military-specific stressor studies							
Binsch et al. [38] (experiment 2)	29 (no info on M or F)	Dutch Armed Forces Air Mobile Brigade School platoon A = 19 ± 1 y Highly physically fit (passed the selection and Air Mobile Brigade fitness test)	**Behavioral operationalization** Successfully completing the Air Mobile Brigade training program	**Week 6:** • Wed, 7 AM (baseline) • Thur, 7 AM (baseline) • Thur, 7:10, 7:20 and 7:30 AM (baseline) **Week 7:** • Wed, 7 AM (baseline) • Thur, 7 AM (baseline) • Thur, 7:10, 7:20 and 7:30 AM (immediately post stressor)	Mann–Whitney U test and regression analyses	Training dropout ($n = 8$), not longer motivated to continue or different expectations Specifically in week 7, the group that persevered in the training had a higher (by 55.0 nmol-min/L) total AUC-cortisol compared with the dropout group The regression equation was significant: Training perseverance = 0.36 + 0.26 · C. $R^2 = 0.12$	In the past five years, course attrition rates ranged from 32 to 68% Main causes of the high attrition rates are described as a 'lack of motivation' and/or that the recruits were 'insufficiently resilient'
Conkright et al. [31]	69 (54 M)	3 Air Force, 56 Army, 6 Marines, and 4 Reserve Officer Training Corps A (M) = 26 ± 5 y A (F) = 26 ± 6 y $\dot{V}O_2$peak (M) = 48 ± 8 mL·kg·min^{-1} $\dot{V}O_2$peak (F) = 41 ± 5 mL·kg·min^{-1}	**Performance operationalization** Change in tactical mobility test performance: • Vertical jump (jump height and maximum force prior to take off) • Water can carry • Fire and movement/casualty drag • 300-m shuttle run • Ruck march Each day at 12:00 h, the tactical mobility test was performed (90 min)	Immediately before and after completing all tactical mobility test events	Pearson's r (r) or Spearman's rho (r_s) were used to determine the relationship between change scores from baseline (day 1) to peak stress (day 3) in tactical mobility test events and biomarker response (i.e., delta before to after tactical mobile test)	The magnitude of release in circulating cortisol was positively correlated with an increase in jump height ($r = 0.413$, $p = 0.005$) A greater decline in IGF-1 response from baseline (day 1) to peak stress (day 3) was associated with an increase in casualty drag time ($r = -0.384$, $p = 0.006$) Change in BDNF response was negatively correlated with changes in maximum force prior to take off for vertical jump ($r_s = -0.377$, $p = 0.020$) and positively correlated with casualty drag time ($r_s = 0.340$, $p = 0.027$)	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Cuddy et al. [37]	11 (no info on M or F)	Candidates striving to become Special Tactics Officers A = 26 ± 3 y	**Behavioral operationalization** Success in Special Tactics Officer selection	Two times per day: • Upon waking (times varied) • ~11:00 am	A mixed design analysis of variance (group × time) with repeated measures for time	There was a trend toward a difference between selected ($N = 4$) and non-selected candidates ($N = 5$) for cortisol, (0.56 ± 0.27 μg·dl^{-1} and 0.89 ± 0.25 μg·dl^{-1}, respectively, $p = 0.09$)	• One subject → unable to pass the initial physical tests • Two subjects → voluntarily dropped out on day 3 • Two subjects → unable to complete the final event due to injury
Farina et al. [54]	800 (M)	Active duty U.S. Army Soldiers Possess an Armed Services Vocational Aptitude Battery General Technical score of 110 or higher	**Behavioral operationalization** Successfully completing the SFAS selection course	Fasted, between 4:30 and 6:30 am, one day preceding the start of selection procedures	Logistic regression	When predictors were analyzed continuously, a one standard deviation increase in cortisol and SHBG predicted that candidates were 1.19 and 1.23 times more likely to be selected, respectively. Those with CRP levels < 9.5 nmol/L were 1.65 times more likely to be selected than those with levels ≥ 9.5 nmol/L	Pearson's correlation coefficient (r) was used to determine the relationship between physiological markers, psychological measures, and physical performance
Farina et al. [62]	233 (M)	Active duty U.S. Army Soldiers A = 25 ± 3 y	**Behavioral operationalization** Successfully completing the SFQC	Fasted, between 4:30 and 6:30 am • One day prior to the start of SFAS • One day following SFAS completion • After completion of SFQC	Univariate logistic regression	Higher post-SFAS cortisol, and greater increases (change) in cortisol from pre-SFAS to post-SFAS predicted the likelihood of graduation from SFQC	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Forse et al. [36]	1006 (79.5% M)	Candidates undergoing a 10-week early career military training program A (M) = 25 ± 3 y A (F) = 25 ± 3 y	**Behavioral operationalization** Successfully completing the US Marine Corps Officer Candidates School	During the first week of training. No further info is provided	Simple logistic regression analyses	Cortisol and DHEA concentration were not significantly associated with attrition	260 candidates (25.8%) attritted over the 10-week training, with the highest number of discharges during week 5
Gepner et al. [42]	20 (M)	Elite combat unit of the Israel Defense Forces A = 20 ± 1 y	**Performance operationalization** Shooting and cognitive performance following five days of field training	> 3 h postprandial, following the final navigational exercise (completed in the morning), right before the cognitive test	Pearson's product-moment correlational analysis	Significant inverse correlations were noted between TNF-α concentrations and dynamic shooting accuracy BDNF concentrations were significantly correlated with the number of correct answers in the cognitive test ($r = 0.672$, $p = 0.012$)	
Hamarsland et al. [40]	15 (no info on M or F)	Only subjects that were able to complete hell week were included A = 23 ± 4 y	**Performance operationalization** Change in CMJ performance and maximal isometric strength in leg press and chest press throughout selection course (7 measurement points)	Morning blood samples (9:00 AM to 10:00 AM) were collected 2 to 3 h after breakfast in each measurement point, except during the measurement point immediately after termination of the hell week and 24 h after	Pearson's correlation coefficients (r) were calculated	Absolute levels and relative changes in IGF-1 tended toward and correlated with the change in CMJ performance during hell week, respectively (absolute: $R = 0.48$, change $R = 0.54$) Similarly, absolute levels and relative changes during hell week for IGFBP3 had a positive correlation with changes in CMJ performance during hell week (absolute: $R = 0.56$, change $R = 0.69$)	Less than 10% are able to complete this course

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Jenz et al. [8]	33 (M)	Airmen candidates that participated in the Battlefield Airmen Preparatory Course A = 22 ± 4 y	**Behavioral operationalization** Successfully completing the subsequent Course of Initial Entry	Between 09:00 and 13:30, but at each day blood draw was carried out within a 30-min interval Five samples were collected: day 1 (start), 3 (after physical exercise), 20 (after extended training), 42 (after extended training) and 52 (after physical exercise)	Mixed-effects ANOVA with days of training being a within factor and graduate status a between factor	14 graduated and 19 did not graduate from the subsequent Course Of Initial Entry Significant main effect differences between grads and non-grads were found for CK, Cl, DHEA-S and TCO_2 Significant effects were also identified for the interaction of graduation status by time (Graduation x Time) for TCO_2, NPY • Day 1 values for NPY were higher in future graduates than in non-graduates • Day 52 values of NPY were lower in future graduates than in non-graduates • Day 52 values of TCO_2 were higher in future graduates than in non-graduates	The goal of the Battlefield Airmen Preparatory Training Course is to increase resilience and to decrease the high rate of attrition that occurs later in the training pipeline
Kalns [58]	122 (no info on M or F)	Candidates of the U.S. Air Force Tactical Air Control Party training A (successful) = 22 ± 4 y A (unsuccessful) = 21 ± 3 y	**Behavioral operationalization** Successfully completing the Tactical Air Control Party training	One saliva sample, taken on the morning of the first day of training before any physical activity was started	T-tests and a classification tree approach	The total number of candidates succeeding is 63 (51.6%) and the number failing is 59 (48.4%) There was no difference in the ESPSLIA/GGHPPPP ratio between the groups that succeed and fail the training The ESPSLIA/GGHPPPP ratio is one of the 4 variables that were necessary to produce a model that predicts success in the training	About 1 in 6 failures occurs because candidates are unable to meet physical requirements. Administrative and academic reasons make up the remaining failures

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Kargl et al. [35]	163 (101 M & 62 F)	College graduates that wish to become US Marine Corps commissioned Officers Candidates enrolled in four different US Marine Corps Officer Candidates School classes between June 2021 March 2023 A = 24 ± 3 y	**Performance operationalization** Performance on the Combat Fitness Test, which takes place during the 4th week of Officer Candidates School	Pre and post Officer Candidates School, fasting state not confirmed, sample collection times varied (primarily before noon)	Pearson's correlation coefficient	Combat fitness test score was negatively correlated with IL 6:IL 10 ratio (r = 0.1916) For the Combat Fitness Test individuals in the low tertile for IL 6 pre, IL 6:IL 10 pre, and CRP post had higher scores	
Ledford et al. [59]	116 (no info on M or F)	SEAL candidates progressing through Navy's Basic Underwater Demolition/SEAL course A = majority < 23 y	**Behavioral operationalization** Successfully completing the First Phase of the Basic Underwater Demolition/SEAL course	First Phase, day 4, between 16:00 and 18:00, non-fasted, post-physical exertion state	ANOVA and regression analysis	54 successfully completed First Phase; 57% of dropped, while the other 43% rolled (for medical or administrative reasons) • DHEA and DHEA/cortisol ratio was higher in successful than in unsuccessful candidates (DHEA: 2.20 and 1.93 ng/mL, $p < 0.01$; DHEA-to-cortisol: 0.211 vs 0.160, $p < 0.001$) • For each one unit increase in the DHEA/cortisol ratio, we expected to see an increase of about 4.3 days of training (β = 0:285)	The attrition rate during Basic Underwater Demolition/SEAL course is approximately 65–80%

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Ledford et al. [23]	380 (no info on M or F)	Active duty men in the U.S. Navy, specifically screened for their athleticism, mental stability, and intelligence A = 23 ± 3 y	**Behavioral operationalization** Successfully completing the complete Basic Underwater Demolition/SEAL course	• Post Basic Orientation (four weeks) • Post Phase 1 (eight weeks) • Post Phase 2 (eight weeks) • Post Phase 3 (eight weeks) Sample collection time varied	Growth models were calculated with Stata's mixed procedure 16.1 (StataCorp, LLC, College Station, TX)	51 ultimately completed Basic Underwater Demolition/SEAL course, while 158 dropped on request, 39 were dropped for performance, and 32 were medically dropped or rolled • The average trajectories for cortisol, BDNF, and NPY were flat • The average candidate experienced a significant growth in DHEA across training (coefficient = 2.32, $p < 0.05$), and the DHEA/cortisol ratio also increased significantly across phases (coefficient = 0.40, $p < 0.001$), with no difference between successful and unsuccessful candidates	Among the 151 successful candidates, 36% completed the course in the class they started, 45% over two classes, 15% over three classes, 4% over four classes, and < 1% over five classes The attrition rate for first-time participants is approximately 70% to 85%
Martin et al. [63]	48 (M)	Midshipmen aspiring to enter the Navy's Sea, Air, and Land program A (range) = 18–26 y Percent body fat = 7.5 ± 2.7%	**Behavioral operationalization** Successfully completing the naval special warfare screener selection course	Fasted, four samples: • Baseline (blood and saliva; 4–6 weeks pre screener) • Post cold pressor test (blood; 4–6 weeks pre screener) • Prescreener (blood and saliva; morning of the screener) • Postscreener (blood and saliva; morning after the screener)	Logistic regression analysis	A decrease in serum cortisol and DHEA from baseline to post cold pressor test in finishers but not in the nonfinishers DHEA/salivary cortisol ratio, DHEA and ACTH increased from pre to postcreener in nonfinishers, while it decreased in finishers GH increased from prescreener to postscreener in the finishers, but no change in the nonfinishers	37 finishers and 11 nonfinishers

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
McFadden et al. [34]	196 (97 M & 99 F)	Civilian recruits	**Performance operationalization** Change in performance on the physical fitness test and combat fitness test during the training	Monday between 04:00 and 05:00 of: • Week 2 • Week 7 • Week 11	Pearson product–moment correlations (r) Change scores were calculated for performance tests Area under the curve was calculated for cortisol responses using trapezoidal method	A weak negative correlation was found between cortisol and changes in physical fitness test score ($p = 0.05$; $r = -0.20$)	23 F and 23 M dropped off or attrited from the USMC throughout the study duration
Morgan III et al. [4]	70 (M)	32 subjects were non-Special Forces, and 38 subjects were Special Forces	**Performance operationalization** Interrogation behavior score	Subgroup 2 (n=21) were sampled at baseline and during exposure to stressor • Stress blood samples were obtained within 12 h of captivity and immediately after exposure to their first military interrogation	Pearson correlation analyses	A significant, positive relationship between the levels of NPY elicited by interrogation stress and the interrogation behavior score ($r = 0.45$; $p < 0.04$)	During the 72-h experience, subjects were permitted to sleep for a total of 19 min
Morgan III et al. [50]	44 (M), subset from Morgan III et al. 2000	23 subjects were Special Forces soldiers and 21 were non-Special Forces soldiers (Rangers and Marines) None had previous combat experience Years in service = 7 ± 2 y	**Performance operationalization** Interrogation behavior score	Subgroup ($n = 21$) were sampled at baseline and during exposure to stressor • Stress blood samples were obtained within 12 h of captivity and immediately after exposure to their first military interrogation	Pearson correlation analyses and stepwise linear regression analyses	Significant positive correlations were observed between interrogation performance scores and percentage of free cortisol in response to interrogation stress ($r = 0.45$, $p < 0.04$) as well as between interrogation performance scores and NPY after interrogation stress ($r = 0.58$, $p < .006$) Stepwise linear regression analyses using interrogation performance scores as the dependent variable and percentage of free cortisol during stress and NPY during stress as the independent variables indicated that the model was significant ($F(1,18) = 9.4$, $p < 0.007$). The model accounted for 31% of the variance (adjusted $R^2 = 0.31$). Percentage of free cortisol during stress had a β value of 0.59 ($t = 3$, $p < .007$). NPY did not make a significant contribution	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Morgan III et al. [51]	25 (No info on M or F)	8 subjects were naval aviators, and 17 were nonaviating marines A = 25 ± 4 y Years in service = 5 ± 4	**Performance operationalization** Interrogation behavior score The performance ratings are scored on a scale that ranges from 0 (no skills demonstrated) to a maximum score of 4 (excellent demonstration of skills)	**Blood and saliva sample** • Baseline: 5 days before stressor exposure at 16:00 • Stress sample: immediately after interrogation, ~16:30 • Recovery sample, 24 h after conclusion of the captivity training, 07:45	Pearson correlation analyses and stepwise linear regression analyses	A significant positive correlation between the DHEA-S/salivary cortisol ratio during stress and the military performance scores ($r = 0.61$; $p = 0.008$) A significant negative correlation between stressor-induced levels of salivary cortisol and military performance ($r = -0.51$; $p = 0.01$) No significant relationships were observed between baseline hormone values, recovery hormone values, DHEA-S/cortisol ratios at baseline or at recovery, and the military performance scores in response to stressor Stepwise linear regression analysis using military performance scores as the dependent variable and the DHEA-S/salivary cortisol ratio and plasma levels of DHEA-S and cortisol during stress as the independent variables showed that the model was significant ($F(1,16) = 5.6$, $p = 0.03$)	
Morgan III et al. [55]	41 (M)	2 Army Rangers, 16 Air Force Pararescue and Combat Control Technician personnel, 4 Marine Reconnaissance personnel, and 19 Army Special Forces soldiers A = 27 ± 4 y	**Performance operationalization** Underwater Navigation Scores The distance from the point of arrival on the beach by the student to the target. Distance was measured in meters and transformed by the training staff into a scaled score of points for candidates	**Blood and saliva sample** • Baseline • After the final examination	Pearson correlation and linear regression analyses	A significant, positive relationship was observed between baseline DHEA as well as baseline DHEA-S and underwater navigation accuracy A positive relationship was observed between post examination DHEA and navigation performance but not between post examination DHEA-S and navigation performance No significant relationships were observed between objective performance and DHEA/cortisol ratios at any of the four time points or between objective performance and DHEA-S/cortisol ratios at any of the four time points Linear regression analyses using baseline values of DHEA-S as the independent variables and underwater navigation performance scores as the dependent variable revealed that model was significant [$F(1,23) = 8.9$; $p < 0.01$]. The β value for DHEA was 0.53 ($t = 2.99$; $p < 0.01$) Linear regression analyses using post-stressor values of DHEA and DHEA-S as the independent variables and underwater navigation as the dependent variable revealed that the model was significant [$F(1,20)$ 7.6; $p < 0.012$]. The β value for DHEA was 0.56 ($t = 2.8$; $p < 0.01$)	Typically, 50% of candidates in a Combat Diver Qualification Course class are eliminated from the course by the end of the first week
Nindl et al. [56]	50 (M)	A = 25 ± 4 y	**Performance operationalization** Change in lower-body power output and maximal lifting strength from pre to post the U.S. Army Ranger course	Morning, between 05:00 and 08:00 Samples were taken before and immediately after the U.S. Army Ranger course	Pearson product–moment correlation analyses	There were no correlations between the changes in hormonal concentrations and the changes in physical performance	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Nindl et al. [32]	122 (29 M & 93 F)	A (M) = 19 ± 1 y A (F) = 19 ± 1 y	**Performance operationalization** Change in $\dot{V}O_2max$ from pre to post training	Two samples: • Pre (within 2 days of reporting to unit) • Post (the day before graduation; 4 months after start)	Pearson product moment correlational analysis	The only significant correlation observed for percent changes was in men between total IGF-1 and $\dot{V}O_2max$ ($r = 0.49$)	A truly gender-integrated basic combat training course
Ojanen et al. [43]	49 (M)	A = 20 ± 1 y	**Performance operationalization** Change in physical performance (same timing as biomarker evaluation): • 12-min running test • Standing long jump • Muscle endurance tests, sit-ups and push-ups, as many repetitions during 60 s	Four samples after an overnight fast between 06:30 and 07:30: • Baseline (7 days prior to start) • Midway (day 13) • End (day 22) • Recovery (day 26)	Bivariate correlation analyses	Changes in absolute and relative IGF-1 concentration between PRE and MID: positively associated with number of push-ups ($r = 0.34$, $p < 0.05$) Changes in absolute and relative CK activity between PRE and MID: negatively associated with the number of push-ups ($r = -0.32$, $p < 0.05$)	
Ojanen et al. [44]	49 (M) same cohort as in Ojanen et al. [43]	A = 20 ± 1 y	**Performance operationalization** Change in upper and lower body strength and shooting accuracy during prolonged Military Field Training (same timing as biomarker evaluation)	Four samples after an overnight fast between 06:30 and 07:30: • Baseline (7 days prior to start) • Midway (day 13) • End (day 22) • Recovery (day 26)	Bivariate correlation analyses	The changes in cortisol and the changes in shooting prone showed a positive correlation in all measurement points ($r = 0.531$, $p \leq 0.001$; $r = 0.337$, $p = 0.024$; $r = 0.572$, $p \leq 0.001$) The changes in IGF-1 correlated negatively ($r = -0.325$, $p = 0.038$) with shooting prone between the PRE and MID measurement points The changes in shooting standing and the changes in testosterone between PRE and POST correlated negatively ($r = -0.378$, $p = 0.010$)	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Pihlainen et al. [45]	49 (M)	Aerobic fitness level higher than 2300 m in the 12-min running test A = 30 ± 9 y	**Performance operationalization** Change in 3000-m running performance (same timing as biomarker evaluation): • High responders (better performance; n = 25) • Low responders (maintain or lower performance; n = 24)	Morning, after a 10-h overnight fast Two samples in Lebanon: • Baseline (day 14) • 5-month follow-up	Mann–Whitney test and Spearman's rank correlation analyses	At baseline, the mean 3000-m test times of the high and low responder group did not differ (866 ± 106 vs 822 ± 85 s) No differences were observed in baseline and relative changes of testosterone, SHBG, IGF-1 and cortisol between groups	This study was also designed as an interventional study: after the baseline measurements, the soldiers were randomly allocated to one of the three combined strength and endurance training groups
Ponce et al. [47]	42 (M)	Second-year cadets in the Military Firefighters Corps A (range) = 22–27 y	**Performance operationalization** Change in countermovement jump performance during training Change in Stroop cognitive performance during training Same timing as biomarker evaluation	Two samples: • Baseline (1 week prior start) • Post (immediately after end of training)	Spearman correlational analyses	Correlational analyses, including both baseline and post values, show that countermovement jump performance was positively correlated with testosterone (rho = 0.69) and IGF-1 (rho = 0.56), and negatively correlated with cortisol (rho = −0.58) and GH (rho = −0.59) Stroop performance was positively correlated with testosterone (rho = 0.41) and IGF-1 (rho = 0.31) and negatively correlated with cortisol (rho = −0.41) and GH (rho = −0.35)	This study did not include an analysis that took into account the time dynamics of the change in biomarkers and performance
Ponce et al. [48]	35 (M)	Brazilian Navy marines A (median, finalists) = 27.6 y A (median, dropouts) = 30 y	**Behavioral operationalization** Successfully completing six operational missions	Fifteen samples (morning): • Baseline • Pre and post each mission	Friedman tests and area-under-the-curve comparisons	No significant differences in CK, creatinine or estimated glomerular filtration rate were found between finalists and dropouts	Five finalists, 30 dropouts

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Sekel et al. [52]	48 (M)	U.S. Military soldiers A = 26 ± 6 y Years of service = 6 ± 5 $\dot{V}O_2$peak = 41 ± 11 $mL \cdot kg \cdot min^{-1}$	**Performance operationalization** Change in military tactical adaptability (SPEAR) performance from day 1 (baseline) to day 3 (peak stress) • Low adaptors (SPEAR score decrease from baseline to peak stress; n = 20) • High adaptors (SPEAR score increase from baseline to peak stress; n = 28)	Fasted, 8:00, each morning of the SMOS training	Two-way mixed-measures ANOVAs	No differences between high and low adaptors in IGF-1, BDNF, cortisol and NPY	High adaptors reported significantly higher scores of self-reported resilience at baseline compared to low adaptors (87.7 ± 9.7 vs. 80.2 ± 11.2, $p = 0.020$, $\eta p2 = 0.119$)
Shumway et al. [60]	Experiment 1 (n = 12; no info on M or F) Experiment 2 (n = 9; no info on M or F)	U.S. Air Force special Warfare trainees **Exp 1** A (average) = 25 y **Exp 2** A (average) = 23 y	**Behavioral operationalization** Passing SIT in Exp 2	**Exp 1** Evening, after academics were complete at 18:00. Exception on last day, samples pulled 2 h after completion of training **Exp 2** Five sporadic blood samples throughout training	Pearson Product-Moment Correlation utilizing a between subjects design Pearson Partial Correlation Controlling for Subject utilizing a within subjects design	In Exp 2, six did not complete the study as they quit, failed, or were medically pulled from training for various reasons None of the biomarkers assessed showed a correlation with passing SIT The two trainees with the highest CK levels (19,859 and 28,453) not only completed SIT, but also recovered back to normal values within 24 h—much quicker than previously annotated in the medical evidence	SIT has a historical pass rate of < 50% despite 1 year of training before this date

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Stein et al. [61]	761 (M) Subset of Farina et al. [54]	Active duty U.S. Army Soldiers enrolled in SFAS A = 25 ± 4 y	**Mixed operationalization** Successfully completing the SFAS selection course Performance on Army Physical Fitness Test, timed runs, timed loaded road marches and land navigation performance	One day preceding start of SFAS, fasted, between 04:30 and 6:30	Principal component analysis Multiple linear regression models with stepwise selection (metabolites = independent variables; physical performance outcomes = dependent variables)	Selected candidates had higher levels of compounds within xenobiotic, pentose phosphate, and corticosteroid metabolic pathways, while non-selected candidates had higher levels of compounds potentially indicative of oxidative stress Higher sphingomyelin, hexanoylcarnitine, 1-carboxyethylphenylalanine, and X-23665 levels were associated with lower performance on load-carriage road marches and run times Higher arabonate/xylonate, X-11315, X-21258, X-25271, and X-25422 were associated with better physical performance on multiple tasks	
Szivak et al. [6]	20 (M)	Active duty men serving in the U.S. Navy and Marine Corps A = 25 ± 4 y	**Performance operationalization** Splitting into high (n = 10) and low (n = 10) fit subgroups based on military physical fitness test scores Change in vertical jump and handgrip tests from Baseline to Stress timepoint	Three testing timepoints: • Baseline: day 1 of SERE training • Stress: day 11 of SERE training • Recovery: 24 h after stress timepoint	Independent T-tests Pearson product-moment correlation analysis	Subjects maintained performance on vertical jump and handgrip tests from Baseline to Stress Physical fitness level did not influence change in plasma adrenaline, dopamine, serum cortisol, or serum testosterone across the SERE training	Women's data were not included due to insufficient n-size ($n = 4$), women's neuroendocrine and physical performance responses trended similarly to those observed for men, except for testosterone

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Tait et al. [33]	46 (37 M & 9 F)	Australian Army recruits A (successful) = 24 ± 7 y A (unsuccessful) = 26 ± 8 y	**Behavioral operationalization** Successfully completing Basic Military Training • 36 completed training on time • 10 did not complete due to backsquadding ($n = 7$) or discharge ($n = 3$)	Weekly: • Upon waking • 30 min post-waking (fasted) • Immediately before bed	Repeated measures ANOVA Generalized linear regression analysis	**Testosterone** No between-group differences for the change over time **Cortisol:** • Unsuccessful candidates had significantly higher waking cortisol at week 5 • Unsuccessful candidates had significantly higher 30-min post-waking cortisol at week 4 & 5 • Unsuccessful candidates had significantly higher bedtime cortisol at week 5 An independent risk factor for unsuccessfulness was higher mean 30 min post-waking cortisol concentration [RR = 1.36 (1.17, 1.59); $p < 0.001$]. Conversely, higher mean testosterone waking levels [RR = 0.98 (0.96–0.99), $p = 0.028$] were associated with decreased risk of unsuccessfulness	
Vaara et al. [11]	69 (no info on M or F)	Finnish conscripts doing compulsory military service A = 19 ± 1 y	**Behavioral operationalization** Successfully completing winter military survival training	One blood sample, fasted, 1 day prior to field training	Logistic regressions and Cox proportional hazard ratios	29% (n = 20) of the conscripts dropped out Only DHEA-S was positively associated with dropout among hormonal variables (β: 0.15, 95% CI: 0.01–0.31, $p = 0.042$) When Cox regressions were applied separately to subgroup of hormonal variables, cortisol predicted dropout (HR: 1.006, 95% CI: 1.0003–1.01, $p = 0.039$) Each 1 nmol/L increase in serum cortisol increased the risk for dropout by 0.06%	
Varanoske et al. [57]	68 (M)	Male U.S. Marines A = 25 ± 3 y	**Performance operationalization** Change in cognitive performance from PRE to POST and REC: • Psychomotor vigilance performance • Grammatical reasoning performance	Three blood samples, overnight fast, between 04:00 and 08:00: **PRE:** day 2 of SERE training **POST:** day 19, immediately after SERE training **REC:** day 45, following 27-day refeeding and recovery period	Mixed model ANOVA Pearson's and Spearman's correlations	Increases in S100B concentrations were associated with decreases in performance on the PVT test (PVT-reaction time: $r = 0.321$, $p = 0.032$; PVT-correct: $\rho = -0.530$, $p < 0.001$)	
Vrijkotte et al. [39]	24 (M)	Male paratrooper soldiers A (completers) = 22 ± 2 y A (dropouts) = 20 ± 3 y	**Behavioral operationalization** Successfully completing an advanced military training course	Four blood samples during three test intervals (i.e., week 0, week 8 and week 12): • Upon arrival in the lab • After first maximal exercise test • Shortly before second maximal exercise test • After second maximal exercise test	Repeated measures ANOVA with the delta scores being the dependent variables and the group the between factor	11 participants completed the course and 13 dropped out No differences found between completers and dropouts for the raw ACTH and cortisol levels at week 0 The delta scores of serum ACTH and cortisol concentration were not found to be significantly different between completers and dropouts at week 0	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Acute military-specific stressor studies							
Goring et al. [53]	115 (M)	Military personnel A = 23 ± 4 y Military experience = 2.5 ± 2 y	**Performance operationalization** Ability to move, shoot, communicate, navigate, and sustain performance under stress	Variable time series between 6 study events	Elastic Net Partial Least Squares Random Forest	None of the analytes emerged as a reliable or consistent predictor across domains	
Iyer et al. [64]	48 (no info on M or F)	Flight cadets A (range) = 18–21 y	**Behavioral operationalization** Successfully completing mid-term tests	Two samples: Immediately pre and post the mid-term tests	Students' T-tests	In passed flight cadets, preflight noradrenaline/adrenaline ratio was lower (3.17 ± 0.45) than in failed flight cadets (5.21 ± 0.98) Passed flight cadets showed an increase in the delta noradrenaline/adrenaline ratio (1.28 ± 0.64), while failed flight cadets showed a reduction (−0.49 ± 0.99)	
Leino et al. [46]	35 (M)	Student pilots A (range) = 19–21 y	**Performance operationalization** Flight performance	Three samples: **PRE:** 15 min before flight **POST:** 5 min after flight **POST2:** 60 min after flight	Pearson correlation analysis	Adrenaline increases correlated with poor flight performance. A high adrenaline level before the flight correlated significantly ($r = 0.43$, $p = 0.02$) with below average results	
Nieto Jimenez et al. [49]	42 (M)	Soldiers going through regular military operations training A = 27 ± 4 y	**Performance operationalization** Time spent in cold water	Three samples at 06:00: **PRE:** 2 days before hypothermia immersion **DURING:** during the induced hypothermia **POST/BASELINE:** low stress; at the end of a 4-month Chilean military operations training during winter	Correlation analysis	Negative correlation between in-water hypothermia time limit and the change (post – during) in ACTH ($r = -0.48$, $p < 0.01$) and cortisol ($r = -0.32$, $p < 0.05$)	

Table 4 (continued)

Study	Sample	Characteristics	Performance resilience outcome	Timing of biomarker evaluation	Statistical analysis	Outcome	Other important information
Shia et al. [65]	17 (M)	Active duty air force members A (median) = 29 y	**Performance operationalization** Change in working memory during the treadmill protocol Change in psychomotor vigilance performance from pre to post treadmill protocol	Four samples, pre, post, 10 min post and 20 min post the treadmill protocol	No info	Cortisol change pre to post positively related to greater increases in working memory accuracy during treadmill protocol ($r = 0.57$, $p = 0.026$) Cortisol change post to 20 min post negatively related to greater increases in working memory accuracy during treadmill protocol ($r = -0.52$, $p = 0.048$) The change in DHEA-S/cortisol from post to 10 min post negatively related to change in working memory reaction time during treadmill protocol ($r = -0.61$, $p = 0.019$) The change in DHEA-S/cortisol from post to 20 min post positively related to change in working memory accuracy during treadmill protocol ($r = 0.70$, $p = 0.011$)	
Vaernes et al. [41]	29 (no info on M or F)	Naval Diving school aspirants	**Performance operationalization** Change in Defense Mechanism Test (diver sketches what he has just seen) performance from pre to during to post dive Change in performance tests (pair-associative recall test, overlearned reasoning tests, and grammar reasoning test) from pre to during to post dive	Samples of blood and urine were taken 5–10 min before and after the dive	No info	Correlation between the Defense Mechanism Test score and cortisol/predive ($r = -0.46$, $p < 0.05$), cortisol/postdive ($r = 0.47$, $p < 0.05$), and prolactine/postdive ($r = -0.45$, $p < 0.01$) Correlations between change in cognitive performance and testosterone/predive (recall capacity at 60 m sea water ($r = 0.40$, $p < 0.05$)), adrenaline/predive (reasoning capacity ($r = -0.60$, $p < 0.01$)), cortisol/postdive (reasoning test at 60 m sea water ($r = -0.38$, $p < 0.05$)), prolactine/postdive (recall capacity at 60 m sea water ($r = 0.63$, $p < 0.01$), GH/postdive (reasoning test $r = -0.50$, $p < 0.05$	

Abbreviations-legend: Age (A), Area Under the Curve (AUC), Adrenocorticotropic hormone (ACTH), Brain-derived neurotrophic factor (BDNF), Countermovement jump (CMJ), Creatine kinase (CK), Chloride (Cl), C-reactive protein (CRP), Dehydroepiandrosterone (DHEA), Dehydroepiandrosterone sulfate (DHEA-S), Dehydroepiandrosterone (sulfate) [DHEA(S)], Female (F), Growth hormone (GH), Insulin-like growth factor-1 (IGF-1), Insulin-like growth factor binding protein (IGFBP), Male (M), Neuropeptide Y (NPY), Navy, Sea, Air, and Land Teams (SEAL), Survival, Evasion, Resistance, and Escape (SERE), Special Forces Assessment and Selection (SFAS), Special Forces Qualification Score (SFQC), Sex hormone binding globulin (SHBG), Stress Inoculation Training (SIT), Simulated military operational stress (SMOS), Total Carbon Dioxide (TCO_2), Tumor Necrosis Factor alpha (TNF-α), Maximal oxygen uptake capacity ($\dot{V}O_2max$), Years (y)

Escape training in which captivity stressors are experienced while undergoing environmental extremes, sleep deprivation, psychological stress, food deprivation and physical demands; and Ledford et al. [23, 59] examined the role of circulating biomarkers in the Basic Underwater Demolition/SEAL course – a 28-week selection process that is designed to ensure that successful candidates meet the minimum mental and physical toughness threshold.

Acute Military-Specific Stressor Studies

Six studies considered the possible role of circulating biomarkers in performance resilience while being exposed to an acute military-specific stressor (see Table 3). Iyer et al. [64] and Leino et al. [46] both included a flight stressor, Nieto Jimenez et al. [49] focused on cold water exposure, Shia et al. [65] physically exhausted participants with a modified Astrand treadmill protocol, Vaernes et al. [41] observed the impact of a dive in a pressure chamber and Goring et al. [53] looked at six different study events that were intended to induce mental and physical stress. The duration of each of these stressors was very similar across studies (i.e., between 40 and 75 min). In terms of intensity and characteristics of the stressor, the acute military stressors are comparable to the prolonged military stressors, insofar as the acute stressors evoked intense physical and/or mental stress. However, whereas the application of acute stressors better enables isolation of one specific aspect of stress, prolonged stressors often combine multiple aspects such as food and sleep deprivation, mental and physical stress, and environmental challenges. Within the acute stressor studies, three studies focused more on the impact of intense mental stressors [41, 46, 64], one study focused on an intense physical stressor [65], one study focused on an environmental challenge (i.e., cold stressor) [49] and one study focused on study events that combine mental and physical stressors [53].

Circulating Biomarkers

Circulating biomarkers of military-specific performance resilience are presented in three different domains, 1) neuroendocrine, 2) anabolic and growth factors and 3) inflammatory. In addition to this classification, we will focus on the timeframe in which biomarker-time dynamics were evaluated.

Prolonged Military Stressor Resilience Studies

As apparent from Table 3, most studies focused on circulating biomarkers belonging to the neuroendocrine and anabolic and growth factor domain. Only 12 of the 34 prolonged military stressor resilience studies also included inflammatory-related circulating biomarkers. Within the neuroendocrine domain, the most prevalent circulating biomarker was cortisol, as included in 26 of the 34 studies. Other frequently included neuroendocrine circulating biomarkers were adrenaline, noradrenaline, NPY and BDNF. Within the anabolic and growth factor domain, Insulin-like growth factor-1 (IGF-1), testosterone, DHEA(S) and growth hormone (GH) were evaluated most. Lastly, the inflammatory biomarkers substances which were analyzed included: Tumor Necrosis Factor Alpha, Interleukin 6, Interleukin 10, C-reactive protein (CRP) and creatine kinase. Interestingly, Stein et al. [61] also investigated a possible role of blood metabolomes in military-specific performance resilience, and Jenz et al. [8], Martin et al. [63], Farina et al. [62] and Shumway et al. [60] examined the most circulating biomarkers.

Table 4 summarizes the timing of the quantification of circulating biomarkers. This overview indicates whether the role of biomarkers was examined via absolute values or via a relative change in time, e.g., pre to post a specific stressor. In both instances it is important to know the timing of the biomarker evaluation relative to the observed stressor. Seven studies [11, 36, 42, 54, 58, 59, 61] were designed in such a way that they included one sample collection and, as such, were limited to evaluating a possible link between circulating biomarkers and performance resilience based on absolute punctual values. Of these seven studies, Farina et al. [54], Kalns et al. [58], Stein et al. [61], Vaara et al. [11] and Forse et al. [36] opted for a baseline collection at the start of a selection course or training, Gepner et al. [42] included a sample collection after five days of field training and right before a military performance task, and Ledford et al. [59] opted for a sample collection in a physically exerted state at the start of a selection course. The other 27 studies included multiple sample collections throughout the selection course, training or operation, providing an opportunity to relate biomarker reactivity to military-specific performance resilience. For example, Conkright et al. [31] sampled blood each day of a five-day Simulated Military Operational Stress (SMOS) training, right before and after completing a tactical mobility performance test at 12:00 h. Jenz et al. [8] collected five blood samples during a 52-day preparatory course. Between 09:00 and 13:30 on day one (baseline), three (after physical exercise), 20 (after extended training), 42 (after extended training) and 52 (after physical exercise). Vrijkotte et al. [39] tested participants at three time points: at training onset, after 8 weeks of paratrooper training, and after completion of 4 weeks of specific commando training. Each test interval participants had to perform the Training OPtimalisation test (TOPtest), that includes two maximal exercise tests on a cycle ergometer and blood was drawn soon as the participants arrived, after the first maximal exercise test, shortly before, and then again after the second maximal exercise test [39]. Morgan III et al. [4, 50] sampled their included participants once at baseline, within 12 h of a captivity stressor, and once during exposure to the stressor, immediately after exposure to captivity stressor. And Martin et al. [63] collected blood samples 4–6 weeks of the screener-stressor (pre and post a cold pressor test), and again pre and post the screener-stressor itself.

Acute Military Stressor Resilience Studies

Table 3 shows that all acute military stressor resilience studies included a neuroendocrine circulating biomarker generally cortisol. Other included neuroendocrine biomarkers were adrenaline and noradrenaline, prolactin and adrenocorticotropic hormone. Four [41, 49, 53, 65] of the six studies also included anabolic and growth factor biomarkers, e.g., testosterone and DHEA(S), while only Goring et al. [53] included an inflammatory biomarker.

Unlike the prolonged military stressor resilience studies, all acute studies sampled multiple times relative to the stressor-exposure (see Table 4 for an overview). All studies included a pre- and a post-stressor sample collection. Additionally, Nieto Jimenez et al. [49] also included a sample collection during a cold water immersion, while Leino et al. [46] and Shia et al. [65] focused on the recovery response by including multiple sample collections post stressor.

Resilience Outcome

Across both the prolonged and acute military stressor resilience studies, most studies – 22 of 40 studies that were included (see Table 4 for more information) – operationalized military-specific performance resilience as the ability to uphold physical and/or cognitive performance while experiencing prolonged and/or acute military-specific stress. Concretely, this most often meant that studies used a physical and/or cognitive performance measure which was completed multiple times throughout a military training or selection course. Subsequently, the change in the performance measure was used to divide the group in a high resilient and low resilient group. The other 18 studies operationalized military-specific performance resilience as successfully completing a military training or selection course (see Table 4 for an overview).

Prolonged Military Stressor Resilience Studies

Circulating Biomarkers and Upholding Military-Specific Performance Ten studies used a physical performance measure to evaluate the ability to uphold performance during a prolonged military training or selection course, i.e., defined as resilience in those studies [6, 31, 32, 34, 35, 40, 43, 45, 56, 61], six studies used a cognitive performance measure [4, 42, 50–52, 57] and three studies used a performance measure that requires both physical and cognitive abilities [44, 47, 55]. In terms of physical performance measures, most studies employed a physical test that assesses lower body strength or a test that evaluates military-specific aerobic performance. For example, a countermovement jump, vertical jump or standing long jump was included as a physical performance measure by Conkright et al. [31], Hamarsland et al. [40], Ojanen et al. [43] and Szivak et al. [6], while Nindl et al. [32], Ojanen et al. [43], Pihlainen et al. [45] and Stein et al. [61] all chose a (loaded) running test to evaluate physical performance resilience. McFadden et al. [34] and Kargl et al. [35] both chose the Combat Fitness Test as a standardized measure of functional/tactical fitness. This test is designed to simulate the physical stressors associated with combat [35]. Regarding cognitive performance measures, computer-based cognitive tests such as a psychomotor vigilance task [57], grammatical reasoning task [57], Stroop task [47] and serial sevens test [42] were included. These tests each evaluate the functioning of a specific cognitive function, for example, sustained attention or response inhibition. In contrast, Morgan III et al. [4, 50, 51] evaluated cognitive function in a more holistic and applied way by creating an “interrogation behavior score”. The score is the sum of observed, classified, target behaviors that survival students were expected to demonstrate and that are expected to enhance a soldier’s success in surviving interrogation by the enemy. Morgan III et al. [55], Ojanen et al. [44] and Ponce et al. [47] also assessed performance in a holistic way and included a performance measure that required both physical and cognitive abilities, e.g., underwater navigation score [55], or included multiple performance measures to evaluate both physical and cognitive performance, e.g., countermovement jump and Stroop performance [47].

An overview of all the links that were observed between the circulating biomarkers and the resilience outcomes can be found in Table 4. In addition, Table 5 presents a visual overview of all the positive and negative associations that were tested and which of them eventually turned out to be statistically significant. In the absolute value columns (i.e., A-columns) a green box indicates that a higher concentration of the biomarker is associated with higher military-specific performance resilience. While in the change score columns (i.e., C-columns) a green box indicates that a higher increase from one timepoint to another is associated with higher military-specific performance resilience. Logically, in both the A- and C-columns, a red box represents the opposite of the green boxes in that same column. In this visual overview we focused on the circulating biomarkers that were defined in

Table 5 Visual representation of the most often dosed circulating biomarkers in the included studies and whether their concentration, absolute value (A) or change score (C), was found to be positively (green) or negatively (red) associated with military-specific performance resilience

	Neuroendocrine										Anabolic and growth factor								Inflammatory									
	Cortisol		Adrenaline		Noradrenaline		NPY		BDNF		IGF-1		Testosterone		DHEA(S)		GH		TNF-α		IL-6		IL-10		CRP		CK	
	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C	A	C
Prolonged military-specific stressor studies																												
Circulating biomarkers of upholding military-specific performance																												
Conkright et al. 2021																												
Gepner et al. 2019																												
Hamarsland et al. 2018																												
Kargl et al. 2024																												
McFadden et al. 2024																												
Morgan III et al. 2000																												
Morgan III et al. 2001																												
Morgan III et al. 2004																												
Morgan III et al. 2009																												
Nindl et al. 2007																												
Nindl et al. 2012																												
Ojanen et al. 2018a																												
Ojanen et al. 2018b																												
Pihlainen et al. 2020																												
Ponce et al. 2023																												
Sekel et al. 2023																												
Szivak et al. 2018																												
Varanoske et al. 2022																												

Table 5 (continued)

Circulating biomarkers of successfully completing a training/selection course																												
Binsch et al. 2017																												
Cuddy et al. 2013																												
Farina et al. 2019																												
Farina et al. 2025																												
Forse et al. 2024																												
Jenz et al. 2021																												
Kalns 2011																												
Ledford et al. 2020																												
Ledford et al. 2022																												
Martin et al. 2025																												
Ponce et al. 2025																												
Shumway et al. 2020																												
Stein et al. 2023																												
Tait et al. 2022																												
Vaara et al. 2020																												
Vrijkotte et al. 2017																												

Table 5 (continued)

Acute military-specific stressor studies																												
Circulating biomarkers of upholding military-specific performance																												
Goring et al. 2025																												
Leino et al. 1999																												
Nieto Jimenez et al. 2018																												
Shia et al. 2015																												
Vaernes et al. 1982																												
Circulating biomarkers of successfully completing a training/selection course																												
Iyer et al. 1994																												

Legend: Absolute value (A), Brain-derived neurotrophic factor (BDNF), Change score (C), Creatine kinase (CK), C-reactive protein (CRP), Dehydroepiandrosterone (sulfate) [DHEA(S)], Growth hormone (GH), Insulin-like growth factor-1 (IGF-1), Interleukin (IL), Neuropeptide Y (NPY), Tumor Necrosis Factor alpha (TNF-α). Grey = this variable was not considered; White = this variable was considered but no link was found or no statistical analysis to evaluate its possible role in resilience was performed; Green = a positive association was reported between the variable and resilience; Red = a negative association was reported between the variable and resilience

section *Circulating biomarkers* as the biomarkers that were most-often evaluated across all included studies. Based on the overview that Table 5 provides, the number of circulating biomarkers that was evaluated in each study was rather limited.

In addition, it is challenging to draw a straightforward conclusion for the most-often measured circulating biomarker (i.e., cortisol). Of the 14 studies which measured cortisol, only one found a positive association between absolute values of cortisol and military performance [50]. Two more studies found a positive association between a change score of cortisol and military performance [31, 44]. Conkright et al. [31] found that the magnitude of release in circulating cortisol was positively correlated with an increase in jump height during a 5-day SMOS protocol. Ojanen et al. [44] in turn showed that changes in cortisol (i.e., an increase) and the changes in shooting prone showed a positive correlation in all measurement points. Unfortunately, no figure or any other information on this correlation is included in the paper, making this correlation difficult to interpret. In contrast to the above mentioned positive associations between cortisol and performance, a negative correlation was observed between absolute cortisol values and military performance in the studies of Morgan III et al. [51], Ponce et al. [47] and McFadden et al. [34]. Morgan III et al. [51] found a negative correlation between stressor-induced levels of salivary cortisol and an instructor-based military performance score. In the study of Ponce et al. [47], a correlational analysis including both absolute baseline and post values showed that countermovement jump performance and Stroop performance were negatively correlated with cortisol; The lower the cortisol value, the higher the vertical jump height or the more Stroop responses per second [47]. In addition, also McFadden et al. [34] found that greater cortisol levels negatively impacted Physical Fitness Test scores.

Other frequently measured biomarkers are IGF-1 and NPY, which had more consistent results across studies of positive correlations. Nine studies included IGF-1 concentration as an outcome measure and six of them found a significant positive association with performance [31, 32, 40, 43, 44, 47]. Four studies included NPY concentration as an outcome measure, and two studies of Morgan III et al. [4, 50] reported positive correlations between the interrogation behavior score and absolute concentrations of NPY post interrogation stressor (i.e., the higher the NPY concentration the better).

The Association of Circulating Biomarkers with Successfully Completing a Training or Selection Course Sixteen studies evaluated the role of circulating biomarkers in successfully completing a military training or selection course

[8, 11, 23, 33, 36–39, 48, 54, 58–63]. All military training and selection courses that were observed were designed to identify candidates who can tolerate a variety of stressors, e.g., sleep and food deprivation, environmental constraints, while maintaining high physical and cognitive function. To attest how difficult it is to successfully complete such military training and/or selection, most studies also refer to the high dropout rates. For example, Binsch et al. [38] focused on the Air Mobile Brigade training program. Graduates of this program will have mastered shooting with standard rifle and machine gun, day/night orienteering, surviving in the field, and executing group and section-level attack and defensive tactics. Course attrition rates range from 32 to 68% [38]. Hamarsland et al. [40] investigated the impact of the annual selection course to join the Norwegian Naval Special Forces, for which less than 10% are able to complete. Ledford et al. [23, 59] included the Basic Underwater Demolition/SEAL course in their study design, for which the attrition rate for first-time participants is approximately 70% to 85% [23].

The visual overview that is provided in Table 5 – similar to the research on the role of circulating biomarkers in upholding military performance – shows that the overlap between studies in circulating biomarkers that were focused upon is limited. Circulating biomarkers in which the most overlap between studies can be found are cortisol, NPY, DHEA(S) and CRP. For cortisol, four [33, 38, 54, 62] out of the 12 studies that measured it found a positive association between cortisol and successfully completing a military training or selection course. All four studies found this positive association with an absolute cortisol value. In contrast, Vaara et al. [11] observed that each 1 nmol/L increase in serum cortisol increased the risk for dropout by 0.06%. Besides the absolute cortisol concentration, three [33, 38, 62] studies also found a positive association between a change score in cortisol and successfully completing a military course. Binsch et al. [38] observed that, specifically in a military challenging condition, the group that persevered in the training had a higher (by 55.0 nmol-min/L) total area-under-the-curve-cortisol value compared with the drop-out group. Tait et al. [33] found, with a repeated measures ANOVA analysis, a significant difference between groups (i.e., unsuccessful vs successful candidates) for the change over time for cortisol levels. At specific timepoints throughout a 12-week Basic Military Training camp, unsuccessful candidates experienced higher waking cortisol, 30-min post waking cortisol and bedtime cortisol levels. Farina et al. [62] observed that greater increases in cortisol from pre to post Special Forces Assessment and Selection predicted the likelihood of graduation from the Special Forces Qualification Course. NPY was measured by Farina et al. [62], Jenz et al. [8], Ledford et al. [23] and Martin et al. [63] but only Jenz et al. [8] observed a relationship between NPY concentrations and graduation status. An interaction effect was observed for NPY concentration, between graduation status and time, indicating that values for NPY were higher in future graduates than in non-graduates on day 1 of the Battlefield Airmen Preparatory Course and, subsequently, were lower on day 52 after performing strenuous physical exercises. For DHEA(S), four [8, 11, 59, 63] out of eight studies found that absolute DHEA(S) concentration was associated with resilience. In the study of Jenz et al. [8] the absolute DHEA-S concentration was a mean calculated from five samples taken over 52 days throughout the Battlefield Airmen Preparatory Course. Four of these five samples were taken following strenuous physical activity and, overall, a higher DHEA-S value was associated with graduation success [8]. The absolute DHEA concentration in Ledford et al. [59] was based on a single sample, taken in a post-physical exertion state, and was found to be higher in successful than unsuccessful candidates. Interestingly, Ledford et al. [59] reported that a higher DHEA/cortisol ratio was also associated with success. Vaara et al. [11] collected one blood sample one day prior to the start of a winter military survival training and concluded that DHEA-S was the only included circulating biomarker that was positively associated with dropout. Lastly, Martin et al. [63] sampled blood before and after a cold pressor test and before and after a 24-h screener stressor and found twice that a decrease in DHEA from pre to post was associated with successfully completing the screener-stressor. The link between CRP and resilience was evaluated by Farina et al. [54, 62], Jenz et al. [8] and Vaara et al. [11]. Farina et al. [54] took one fasted blood sample one day before the start of selection procedures and found that those with CRP levels below 9.5 nmol/L were more likely to be selected. In another study, Farina et al. [62] did however not find any role for CRP levels. Vaara et al. [11] also took one blood sample prior to the start of field training and Jenz et al. [8] took five samples over 52 days throughout the Battlefield Airmen Preparatory Course, but both did not find an association between CRP levels and resilience.

Acute Military Stressor Resilience Studies

Circulating Biomarkers and Upholding Military-Specific Performance Five [41, 46, 49, 53, 65] of the six acute military stressor resilience studies looked at the role of circulating biomarkers in upholding military-specific cognitive and/or physical performance. Leino et al. [46] focused on flight performance, while Nieto Jimenez et al. [49] used time spent in cold water as a performance measure. Shia et al. [65] and Vaernes et al. [41] both looked at specific cognitive aspects to evaluate resilience in an acute military stressor scenario. Shia et al. [65] focused on working memory and psychomotor vigilance during a treadmill protocol, while Vaernes et al. [41] let divers make sketches of what they had just seen before, during and immediately after a dive (i.e., Defense Mechanism Test). Lastly, Goring et al. [53] employed multiple outcome measures (see Brunyé et al. [66]) to score participants on military-specific performance in terms of moving, shooting, communicating, navigating and sustaining while being exposed to military-specific stressors. For example, participants are exposed to an acute physical stressor (threat of torso shock) while attempting to perform difficult marksmanship, memory, and spatial orienting tasks in virtual reality.

All five studies looked at cortisol and three found a significant association [41, 49, 65]. Nieto Jimenez et al. [49] reported that cortisol increases more from during to post cold-water exposure in subjects who tolerate less time in cold water. Vaernes et al. [41] found correlations between absolute pre-dive and post-dive cortisol concentrations on the one hand and cognitive performance and the change in cognitive performance on the other hand. Shia et al. [65] put forward that the cortisol change from pre to post was positively related to greater increases in working memory accuracy during the treadmill protocol. On top of that, Shia et al. [65] found positive and negative correlations between acute recovery changes, up until 20 min post the treadmill protocol, in cortisol and DHEA-S/cortisol ratio and working memory performance.

The Association Between Circulating Biomarkers and Successfully Completing a Training or Selection Course Iyer et al. [64] evaluated a possible role of adrenaline and noradrenaline in flight cadets successfully completing mid-term tests. To do so they collected a blood sample immediately pre and post the mid-term tests. In passed flight cadets they found that preflight noradrenaline/adrenaline ratio was lower than in failed flight cadets [64]. Moreover, passed flight cadets also showed an increase in the noradrenaline/adrenaline ratio, while failed flight cadets showed a reduction [64].

Discussion

In this review, we sought to systematically summarize and review the available knowledge on circulating biomarkers of military-specific performance resilience. All studies but one that are included in this review were of moderate to strong quality based on the QualSyst-tool. Most studies lost points for the description of the participant characteristics and controlling for confounding variables. This observation highlights a specific difficulty in this field of research, namely the sensitivity of military data, resulting in poor description of participants' characteristics. Consequently, poor description of the sample may hinder adequate control of potential confounding variables, such as age, sex, military experience, and physical fitness level. A total of 40 studies were included. Of these, 34 included a prolonged military-specific stressor in their study design (24 h to two years), while the other six studies looked at biomarkers reflecting performance resilience while being exposed to an acute military-specific stressor (40 min to 75 min). In addition, 22 of the 40 included studies operationalized military-specific performance resilience as the ability to uphold physical and/or cognitive performance while experiencing prolonged and/or acute military-specific stress, which allows for a more standardized experimental approach; whereas the other 18 studies operationalized military-specific performance resilience as successfully completing a military training or selection course, which is a more ecologically valid result. Half of the studies that measured cortisol (15/31 studies), NPY (3/8 studies), IGF-1 (6/11 studies) and DHEA(S) (5/13 studies) reported an association with military-specific performance resilience. In general, a more pronounced but controlled cortisol response to acute stressors, and rapid recovery to baseline values seems to reflect military-specific resilience. Alongside this stressor-induced cortisol response, a corresponding release of NPY and DHEA(S) is triggered, and both are seen as a recovery biomarker. Similarly, as with the cortisol response, both the NPY and DHEA(S) responses need to be pronounced but controlled and rapidly recover to baseline levels to reflect the ability to tolerate stress and maintain engagement and performance in a stressful environment.

This conclusion aligns with the theoretical physiological response of the human body described in the introduction. It further suggests that, rather than attempting to define absolute thresholds for cortisol, NPY and DHEA(S) levels, the true value of screening and monitoring circulating biomarkers lies in assessing relative thresholds, e.g., percentage changes from individual baselines. Relative thresholds will provide the ability to interpret whether a pronounced but controlled stressor-related increase in cortisol, NPY, and DHEA(S) is present, and whether they rapidly recover to baseline levels. Working with relative thresholds accounts for the high interindividual variability in circulating biomarkers (i.e., baseline levels vary widely between individuals due to genetics, sex and training status). Moreover, determining these relative thresholds in relation to a physical stressor, e.g., a specific physical performance task that is performed multiple times throughout a selection course, filters out the impact of multiple other military-specific stressors that are also known to impact the level of circulating biomarkers, including sleep deprivation, energy deficits, variable environmental conditions, and emotional and psychological strain.

Study and Sample Characteristics

The review by Beckner et al. [20] posited that prior military experience, or lack thereof, is an important factor to consider comparing military resilience research results. The present review provides an overview of the prior military experience of each study cohort to take this factor into account. Unfortunately, only four studies [50–53] reported the number of years of service of the study participants, i.e., three of them reported ~6 years of service, making it impossible to quantify this factor. Of course, besides military experience, multiple other factors play a role in the observed effects on circulating biomarker responses. Some of those factors, e.g., age, military stressor paradigms, physical and mental fitness, have been reported by most/all studies and an overview is provided in Tables 3 and 4. In terms of military stressor paradigms there is, as already mentioned by Beckner et al. [20], a wide range of paradigms from basic military training to elite forces selection courses. The main trade-off regarding this operationalization is standardization versus ecological validity. The average age of the participants ranged from 19 to 30 years and regarding physical and mental fitness, most studies were comparable and included participants that had to demonstrate high physical and/or mental fitness before being included and/or had to successfully complete a certain preparatory course first (see Table 4). However, although most/all studies included information on age and physical and mental fitness, this information was often not provided in detail and seldomly used to be included as a covariate and to be controlled for as a confounding variable (see Table 2).

Another crucial factor when interpreting circulating biomarkers responses is sex. The fact that on a total research population of 5058 participants only 484 of them are women (i.e., some studies did not include sex information on the included cohort and, therefore, it is unknown whether those studies included women) underlines that women are still underrepresented in applied resilience research, thus this field is heavily skewed in this regard. Six different studies included women [31–36], and only three used that opportunity to assess whether sex impacted the role of biomarkers in resilience [31–33]. Tait et al. [33] included 37 men and nine women and reported that dropping out of a Basic Military Training was linked with higher mean 30 min post-waking cortisol concentration. Conversely, higher mean testosterone waking levels were associated with decreased risk of dropping out [33]. For both findings, Tait et al. [33] did not observe sex to be a significant covariate. In contrast, Conkright et al. [31] and Nindl et al. [32] did report an impact of sex. Nindl et al. [32] included 29 men and 93 women and observed that, during sex-integrated Israeli Army Basic Combat Training, men and women generally respond in a similar fashion with regard to the IGF-1 system and inflammatory cytokines. However, the only significant correlations between circulating biomarkers and resilience were observed for the positive percent change in the total IGF-1 in men, which was positively correlated with the positive percent change in $\dot{V}O_2$max [32]. Conkright et al. [31] included 54 men and 15 women and found that the pattern of change of GH, IGF-1, cortisol and BDNF from before to after a physically exerting protocol was significantly different between sexes. The physical exerting protocol was performed each day during a five-day SMOS paradigm [31]. Interestingly, Conkright et al. [31] also collected menstrual cycle and contraception data and reported that 58% of the women reported using a form of birth control. Due to lack of statistical power Conkright et al. [31] did not perform statistical analysis that took this information into account. Nevertheless, it sets a good example, and it provides important background information for future work.

Circulating Biomarkers of Military-Specific Performance Resilience

As already mentioned in the beginning of this discussion, cortisol and NPY are the most frequently evaluated neuroendocrine circulating biomarkers, while IGF-1 and DHEA(S) are the most frequently evaluated anabolic and growth factor biomarkers.

In terms of cortisol, conflicting results were observed. Morgan III et al. [50] reported that higher post-stressor cortisol is related to better performance, whereas Morgan III et al. [51], Ponce et al. [47] and McFadden et al. [34] found that lower pre- and post-stressor cortisol is related to better performance (see Tables 3, 4 and 5). These conflicting results can partially be explained by several factors, e.g., sampling moment relative to stressor, time of day, prior exposure to stressors. The research on circulating biomarkers in successfully completing a training or selection course (see Section "The association of circulating biomarkers with successfully completing a training or selection course") provides some insight in this matter. For example, the study of Tait et al. [33] neatly demonstrated the added value of timing the sampling relative to wake- and bedtime. Higher concentrations of morning, i.e., immediately and 30 min post waking, and bedtime cortisol across weeks 1 to 6 of a 12-weeks Basic Military Training course were found in individuals that eventually failed compared to those who succeeded [33]. This suggests that a persistent stress response can occur in unsuccessful candidates and that increased cortisol levels can be associated with dropping out if this increase is unrelated to an acute stressor (i.e., wake- and bedtime). These findings underscore the need to account for time-of-day effects and to consider integrated measures such as the area under the curve when interpreting cortisol data, to distinguish baseline differences from acute responses to stressors. In addition to the study of Tait et al. [33], Vaara et al. [11] timed their blood sampling relative to the start of a winter survival training and found a negative correlation between cortisol values and completion success, i.e., higher cortisol values were related to dropout. In that sense, increased absolute cortisol values were interpreted as a result of intensive physical preparation training, mental anticipation prior to the field training or their combination [11]. Subsequently, Vaara et al. [11] speculated that the higher cortisol levels at rest reflected a sign of overreaching or overtraining. However, the likelihood that this speculation is true is very low. Previous research already demonstrated that to diagnose overreaching and/or overtraining it is important to synchronize their sampling in accordance with their response patterns [67]. Emphasizing the importance of taking multiple samples to enable the calculation of a percent change and the determination of a response pattern.

Conflicting results in changes in cortisol through acute military stressors are observed, which can also be explained through the differences in timing of collection. For example, conflicting results were observed in two acute military stressor resilience studies assessing the association between change in cortisol and performance resilience: Nieto Jimenez et al. [49] observed that a higher increase in cortisol from during to post cold water exposure was negatively related to maximal time in cold water, whereas, Shia et al. [65] observed a positive association between an increase in cortisol from baseline and increases in working memory accuracy immediately following an exhaustive treadmill protocol. However, the relationship observed by Shia et al. [65] was reversed 20 min later, with those still showing elevated cortisol levels also showing decreases in working memory accuracy. Also important to notice here is that the 'during measurement' of Nieto Jimenez et al. [49] was taken immediately after exiting the cold water and the 'post measurement' was taken at a later stage, at the end of a 4-month Chilean military operations training. In that sense, the study of Shia et al. [65] provides more insight with a blood sampling design that was tightly scheduled around the acute treadmill stressor. Based on their results, Shia et al. [65] put forward that quick and high spikes in cortisol during exposure to stressors, along with a group of catecholamines such as adrenaline, noradrenaline, and dopamine coupled with rapid declines back to baseline levels is characteristic of physiological toughness or stress resilience. This interpretation is supported by research on performance resilience in prolonged military-specific stressor studies [31, 44]. Overall, acute changes in cortisol concentration, like in any other circulating biomarker, should be interpreted relatively to a baseline concentration that is taken closely to the stressor exposure. Comparing a post stressor measurement to a pre measurement that is taken closely to stressor exposure controls for the impact of other military-specific (e.g., sleep deprivation) and non-military specific (e.g., circadian rhythm) stressors on the observed circulating biomarker. Therefore, the pre- and post-stressor measurements that are necessary to quantify an acute change, should be taken as close as possible to the moment of stressor exposure. Additionally, time of day must be carefully considered, as many biomarkers exhibit circadian variation that can influence observed responses.

For NPY the findings are, despite multiple null findings, rather consistent (see Tables 4 and 5). This is not surprising, as NPY is far less versatile than cortisol when considering normal physiological functioning and might thus be a more adequate candidate to reflect specific resilience responses. Both in the research of Morgan III et al. [4, 50] and Jenz et al. [8] a more pronounced NPY response to an acute stressor, and rapid recovery to baseline NPY values (i.e., within 24 h) have been associated with, respectively, better interrogation behavior scores and graduation success on a Course Of Initial Entry. Interestingly, in the studies of Morgan III et al. [4, 50] blood was sampled during a 72 h army survival training course, while Jenz et al. [8] sampled blood five times throughout a 52-day Battlefield Airmen Preparatory Course. The fact that in both studies 1) higher NPY levels during, or right after, acute stressors and 2) more rapid return to baseline NPY levels once the recovery period starts, are related to military-specific performance resilience demonstrates that NPY reactivity can be of value both in acute and prolonged stress situations. Unfortunately, no acute military stressor resilience study that was included in the present review included an NPY measurement and, therefore, this cannot be further confirmed by their results. In addition to interpreting NPY values as a stand-alone biomarker, the results of Morgan III et al. [50] also showed the importance of interpreting circulating biomarker values relatively to each other. The peripheral change in cortisol and NPY (i.e., an increase) were highly correlated, and Morgan III et al. [50] suggested that stressor-induced release of cortisol elicited a corresponding release of NPY (i.e., an anxiolytic agent). Moreover, Morgan III et al. [50] also suggested that certain individuals, as they encounter stress, reach a threshold at which they are unable to counter-regulate their response to a stressor with an adequate release of NPY. This inadequacy is thought to lead to the inability to tolerate stress, dissociation from the environment, and eventually decreased military-specific performance.

Similarly to NPY, DHEA(S) (i.e., a precursor anabolic steroid) is seen as a recovery biomarker—memory enhancing, antidepressant, anxiolytic, and anti-aggression properties [51]—that is activated during stress. Subsequently, DHEA(S) can be interpreted as a stand-alone biomarker, but is also often interpreted relative to cortisol values (see Shia et al. [65]). In terms of the relation between post physical exertion-DHEA(S) concentration and performance resilience, all reported findings suggest that a higher post-exertion DHEA(S) concentration is positively related to performance resilience [8, 55, 59]. In contrast, in terms of the link between pre-stressor DHEA(S) concentration and performance resilience, there is less unity. Morgan III et al. [55] reported a positive relationship, i.e., higher baseline DHEA levels may buffer against the effects of prolonged stress and promote performance, while Vaara et al. [11] found a negative relationship. No interpretation of this result was given by Vaara et al. [11]. However, unlike the discrepancy in cortisol, this discrepancy does not seem to be explained by differing sampling time points, as subjects in both studies were sampled the day before the start of the military course. Though speculative, the difference could be related to the differing background of the participants in both studies. Morgan III et al. [55] collected samples during a Special Forces Diver Qualification Course in which the participants were on average 27 years old, while Vaara et al. [11] included Finnish conscripts doing compulsory military service at the age of 19. This differing background could impact the way how basal concentrations must be interpreted, e.g., in less experienced individuals, higher basal concentrations may be reflective of impaired recovery of activities performed in preparation of a military course or reflect anticipatory stress [54]. The idea that background impacts the way how basal concentrations must be interpreted is further strengthened by the fact that, compared with the normal range of DHEA in adults (approximately 1.33–7.78 ng/mL for individuals aged 18–40 years [68]), the reported DHEA concentrations in both studies fall within the normal range but at opposite ends. Specifically, Morgan III et al. [55] reported a mean DHEA concentration of 7.3 ng/mL, corresponding to the upper end of the normal range, whereas Vaara et al. [11] reported a mean concentration of 2.7 ng/mL, near the lower end.

Interpreting DHEA(S) concentration relative to cortisol values has also provided value to performance resilience [23, 51, 55, 59, 63, 65], as it represents a surrogate for the relative proportion of anabolic to catabolic hormones, i.e., DHEA counters the catabolic effects of sustained elevated cortisol. Despite one study finding the opposite [63], several studies consistently show that a higher DHEA(S)/Cortisol ratio is related to better performance resilience [23, 51, 59]. Although a follow-up study [55] failed to replicate these findings, this again was possibly due to a larger time delay in hormone sampling after cessation of the acute stressor (45 min [55] vs 5 to 10 min [51]). Shia et al. [65] provided additional evidence for a possible role of DHEA-S in a rapid decline of cortisol after the initial cortisol spike during exposure to a stressor.

Regarding IGF-1, unlike the other evaluated biomarkers, most studies focused on the relative change in IGF-1 over time instead of an absolute value (see Table 5). From the ten studies that included IGF-1 in their study design, nine evaluated a possible link between IGF-1 and military-specific performance throughout prolonged military stress. Six of those nine studies observed a positive link between IGF-1 and performance resilience (see Table 5). For example, Conkright et al. [31] sampled blood immediately before and after a 90-min tactical mobility test protocol that was performed each day during a five-day SMOS protocol and found that a greater decline in IGF-I response from baseline (day 1) to peak stress (day 3) was associated with an increase in casualty drag time (i.e., one of the tests performed during the tactical mobility test protocol). The positive association between IGF-1 and upholding performance during exposure to a prolonged stressor could not be translated to the aspects of course completion., as Vaara et al. [11] did not find a link between IGF-1 concentration and successfully completing a 10-day winter survival training. However, this could be explained by the study design in which only a baseline measure was taken one day before the start of the winter survival training. Overall, these support a positive relationship between IGF-1 and military performance resilience.

While most of the above-mentioned studies measured biomarker concentrations in blood, some studies used saliva to determine biomarker concentration (see Table 3 for an overview). It is important to note that differences in biofluid type (blood vs saliva) and analytical methods (ELISA vs mass spectrometry; see Table 3 for an overview) may also contribute to variability in reported results. For example, cortisol measured in blood reflects total circulating hormone, including protein-bound fractions, whereas salivary cortisol reflects the biologically active free fraction, which may respond more rapidly to an acute stressor [69]. Additionally, analytical techniques such as ELISA and mass spectrometry differ in specificity and sensitivity, with ELISA being more prone to cross-reactivity [70] and mass spectrometry providing greater accuracy. These methodological differences should also be considered when interpreting conflicting findings across studies.

Potential Military Applications

There are several potential military applications from this field of research. A first one suggests that hormonal signaling often precedes phenotypic outcomes [8, 31, 47]. Our review clearly shows that hormonal changes are associated with physical performance changes and with military-specific performance resilience. Consequently, although screening and monitoring for stressor-related hormonal changes present financial and logistical challenges, such measures may be useful not only as early warning indicators of subsequent declines in physical performance and performance resilience, e.g., both in selection courses as well as in operational soldiers, but also to provide insight into how soldiers' physiological characteristic are impacted by stressors. This insight can be used to sharpen soldiers' self-management and design interventions and training methods that positively impact their functioning and health both on the short term (e.g., during a mission) and the long term (e.g., the longevity of their military career). In light of this potential military application, others [40] have emphasized that the monitoring of biomarker changes *per* se has its limitations in predicting resilience, for example it is insensitive to the post-stressor recovery of physical performance in soldiers. It is therefore important that the changes are interpreted in light of each other, and in light of psychological (e.g., mood questionnaires) and behavioral measures (e.g., mission performance). Considering the current gaps in knowledge—we measure statistical associations, not mechanistical causalities—it seems reasonable to cast a wider net than solely biological markers, e.g., physiological load monitoring with actigraphs, psychometric assessments. See the publication of Koltun et al. [71] for a review on this matter.

A second military application is a logical consequence of the first application, being that the screening and monitoring of biomarker changes could eventually also be used to detect clinical pathologies such as overtraining and PTSD. For example, individual variation in neuroendocrine responses exist before trauma exposure, and may be used to test constructs of vulnerability to develop PTSD in humans [50]. Besides providing valuable input in the prediction of drops in performance resilience and the occurrence of clinical stress-related pathologies (e.g., PTSD), the screening and monitoring of hormonal changes could also be used to finetune the content and structure of multiple military (selection) courses. Depending on the goal of the military course, 1) training participants to handle specific circumstances and stressors or 2) designing a course to select in those individuals that have what it takes to become part of a specific unit, the sequence of stressors (e.g., food deprivation, sleep deprivation, heat, captivity stress) could be adapted to optimize the military course outcomes and avoid unnecessary attrition. For example, the sequence of military stressors could be adapted in such a way, e.g., sufficient recovery, that training adaptations are optimally triggered, or it could be adapted so that military-specific stressors are applied in the most ecological way and the selection process is as efficient as possible.

The above-described military applications are also applicable to other populations. Developing self-management techniques, interventions and training methods to optimize performance resilience are of use to a wide array of populations that aim to perform in a stressful environment, e.g., athletes, police officers, emergency rescue workers, medical personnel. In addition, the detection of stress-related clinical pathologies such as PTSD has, of course, also added value to clinical populations outside of the military community.

Future Directions

Outside of the military context, the quest for the physiological basis of resilience has brought Seligowski et al. [72] to the conclusion that resilience should be conceptualized as a trait. Taken together, the research they reviewed suggests that several candidate genes and Single Nucleotide Polymorphisms, along with physiological mechanisms and neuroimaging characteristics, provide a biological model of resilience pathways [72]. For future resilience research Seligowski et al. [72] advised to study resilience as a trait and to examine biological markers in association with specific resilience measures, such as the Connor-Davidson Resilience Scale (i.e., a trait resilience questionnaire) [72]. However, in contrast to the trait perspective, the process perspective suggests that coping with stress builds resilience through learning and memory mechanisms [73].

Similarly, resilience can also shift negatively and decrease in time. For example, optimism – one of the six core psychosocial factors that was associated with resilience [74] – can shift over time or across situations and, introduces a state facet in the construct resilience. For future military research, both the trait and state facets of resilience should be considered when designing resilience studies. Investigations should aim to create insight in the role of state factors, e.g., age, body mass, physique and composition, aerobic capacity, previous stress exposure, illness, medication, mood, nutritional, and hydration status, in inter- and intra-individual differences in the acute stressor reactivity of circulating biomarkers.

Recent studies [75, 76] evaluating whether resilience training modulates the human response to a stressor are good examples of such state-related studies. Additional studies of this type should be performed to, for example, determine a possible role of physical fitness. On top of state factors impacting resilience, stable differences between individuals in the reactivity of different circulating biomarkers in response to stressors exist, which need to be considered as well. For example, Chang et al. [77] demonstrated that a functional promoter polymorphism of the NPY gene alters chronic stress-related vagal control, confirming that resilience-associated responses to a stressor are, at least in part, a trait. In addition, both in the state and trait resilience research, it will be of added value to not only look at resilience on an individual, but also on a team level, As illustrated by Bowers et al. [78], who provide a theoretical model that can be used by future research to consider team resilience. This state vs trait approach is a contemporary version of the nature vs nurture question, where both aspects need to be taken into account, as they most probably interact.

The screening and monitoring of changes in circulating biomarkers in the field requires the development of a wearable device. The development of wearables for cortisol detection is extremely popular [22, 79, 80]. Specifically related to the military context, Flatebo et al. [80] recently published a review in which the electrochemical aptamer-based sensor platform was put forward as a promising avenue to make continuous, real-time molecular measurements become available in a platform technology in the future.

Eventually, the availability of a wearable device to monitor circulating biomarkers will create the necessity to standardize measurement timepoints to make the monitoring values interpretable and translatable to the individual wearing the device. Future military-specific research on this aspect can benefit from the advancements that have been made in the field of sports research. For example, Crewther et al. [81] reported in detail how they assessed the awakening responses of cortisol and testosterone in male athletes across light and heavy training weeks. In military research, researchers often need to be flexible in their study designs, i.e., adapt to military courses and unforeseen circumstances, to get the opportunity to even collect data. Nonetheless, in the future, a detailed, rigid and well-thought-out methodology in which diurnal variations are considered and blood sampling timepoints are ideally timed in light of a specific stressor—e.g., The TOPtest [39] or the cold pressor test [63]—or the moment of awakening will be necessary to create additional insights and optimize between-study comparability and interpretability in circulating biomarker values.

For example, in the study of Vrijkotte et al. [39] a stressor paradigm was designed that included two maximal exercise tests in one day. To quantify the stressor reactivity, blood was drawn at four moments throughout this paradigm: as soon as the participants arrived, after the first maximal exercise test, shortly before, and then again after the second maximal exercise test. Such a design enables one to evaluate the robustness of the acute response to a stressor, i.e., 1) pre to post first maximal exercise test and 2) pre to post second maximal exercise test, and, in addition, also provides the ability to interpret whether a rapid return to baseline levels is present, i.e., comparison between pre first maximal exercise test and pre second maximal exercise test. The study of Vrijkotte et al. [39] is therefore a good example of how a study should be designed to determine whether a response to a stressor deviates from the normal response (see 1. Introduction) and to evaluate whether an acute response to a stressor is related to performance resilience.

Moreover, the TOPtest itself, and more specifically the adrenocorticotrophic hormone and prolactin responses to the second exercise test, have been shown to discriminate between nonfunctional overreaching and overtraining syndrome [82]. This shows that it is possible to define cutoff thresholds for hormonal responses to acute stressors to define overtraining status and provides a proof of principle to further develop similar cutoff thresholds for military-specific performance resilience. Importantly, Buyse et al. [82] emphasized in their paper that, despite the adrenocorticotrophic hormone and prolactin responses to the second exercise test in the TOP test are most indicative for the diagnosis of nonfunctional overreaching and overtraining syndrome—>200% for nonfunctional overreaching and < 200% for overtraining syndrome – their sensitivity is too low to use these values as biomarkers for nonfunctional overreaching and overtraining syndrome, without taking into account other psychological and physiological factors. This highlights that, for assessing military-specific performance resilience, a holistic approach is essential, one that integrates both physiological and psychological components. Evaluating responses to different types of stressors is necessary, as individuals' reactions vary depending on factors such as physiology and previous experiences.

Conclusion

A total of 40 studies on the topic were included in this systematic review, and we present evidence showing that circulating biomarkers of military-specific performance resilience are relevant. Changes in neuroendocrine (i.e., cortisol and NPY) and anabolic and growth factors [i.e., IGF-1 and DHEA(S)] are linked with an individual's ability to uphold military-specific performance (i.e., both cognitive and physical performance) and with the graduation success in military trainings/selection courses. Screening and monitoring of their reactivity in response to an acute stressor, both in an acute stressor and a prolonged stressor scenario, provide a physiological window into military-specific performance resilience. In general, a more pronounced but controlled cortisol response to an acute stressor and rapid recovery to baseline values promotes military-specific resilience. Alongside this stressor-induced cortisol response, a corresponding release of NPY and DHEA(S) is triggered, and both are seen as recovery biomarkers. Similarly, as with the cortisol response, both the NPY and DHEA(S) responses need to be pronounced and rapidly recover to baseline levels to promote the ability to tolerate stress and maintain engagement and performance in a stressful environment. One of the challenges for future research will be to determine relative thresholds for the stressor-related in- or decrease in circulating biomarker levels and relative timing to stressor to correctly interpret 'a pronounced but controlled response' and 'rapid recovery to baseline'. Currently (and unfortunately) this field of research is immature and the inter-study differences in methodology are too broad to define such relative thresholds. In addition, the results of the present review suggest a role for state-related factors such as age and prior exposure to (military) stressors. Future research defining the state – which can be changed through training or behavior—and/or trait – which can be applied for selection—aspects of the military-specific physiological response to a stressor would be highly valuable. Eventually, this field of research on circulatory biomarkers will open new opportunities to improve military-specific resilience, readiness and health by developing wearables that can improve self-management of soldiers via screening and monitoring (e.g., resilience training and/or the application of countermeasures based on screening and/or monitoring).

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interest The authors declare no competing interests.

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